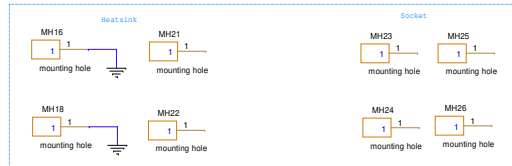


AGILEX™ 5 FPGA E-SERIES 065A PREMIUM DEVELOPMENT KIT (PRODUCTION)

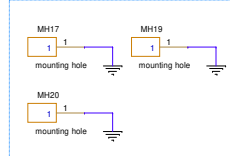
REV	DATE	PAGES	DESCRIPTION
A1	05/30/2024	All	Initial Release
A2	09/23/2024	All	Update to Altera logo, and Altera copy right declaration
A3	06/23/2025	13-16 8, 16, 19, 44	Update CLK_SEL, UART0 default state to Low status Update IO98B CDR assignment per A5ED065B pinout Clean up unused circuitry
	09/17/2025	All	Dev Kit Production Version, Update Altera Schematic Template

PAGE	DESCRIPTION	PAGE	DESCRIPTION
1	Cover, Table of Contents, Notes	38	SFP1
2	Block Diagram	39	SFP2
3	Power Tree Diagram	40	XCVR SMAs
4	Power Sequence Diagram	41	FMC+ PIN TABLE
5	Clock Diagram	42	FMC+ CONNECTOR
6	I2C Diagram	43	10/100/1G/2.5G SGMII ENET1
7	JTAG Diagram	44	BLANK
8	MAX10 SYS Control 1/2	45	10/100/1G RGMII ENET
9	MAX10 SYS Control 2/2	46	USB Diagram
10	AVST Config Flash & HyperRAM	47	USB HUB, BLASTER, UART
11	SDM Configuration	48	FPGA SWITCHES,BUTTONS
12	HPS IO48	49	SYSTEM LEDS, FAN
13	HSIO 2ALR - LPDDR4	50	I2C DEVICES, NIOS FLASH
14	HSIO 2BLR - COMP1	51	LEVEL TRANSLATORS
15	HSIO 3ALR - DDR4 COMP_HPS	52	CLOCKS 1 - PTP 1588
16	HSIO 3BLR - FMC/MIPI/SGMII	53	CLOCKS 2 - SI5532
17	HVIO 5AB - QSFP,ENET	54	12V INPUT & 12V_EARLY
18	HVIO 6AB - BTN,SW,SFP	55	5V0,3V3_SYS, 3V3, 2V5, 1V8,1V2_MAX
19	HVIO 6CD - FMC+,RGMII	56	12V_SYS
20	XCVR 1ABC - QSFP/USB3/SGMII	57	AD17883 CONTROLLER1 1/2
21	XCVR 4ABC - SFP/FMC+	58	AD17883 CONTROLLER1 2/2
22	FPGA DNU & NC	59	VCCcore PHASE1
23	FPGA POWER	60	VCCcore PHASE2
24	FPGA GND	61	VCC_HSSI, VCCERT_UX
25	FPGA DECAPS	62	AD17883 CONTROLLER2 1/2
26	LPDDR4 COMPONENT	63	AD17883 CONTROLLER2 2/2
27	DDR4 COMP1 BYTE 0-1	64	VCCOPT, VCCIO_3B_B
28	DDR4 COMP1 BYTE 2-3	65	HSIO_2B3A3B_T, VCCIO_HVIO
29	DDR4 COMP1 ECC	66	VCCIO_2A, 2V5_SYS, 1V8LP4
30	HPS DDR4 COMP BYTE 0-1	67	DDR4 VREF, VTT
31	HPS DDR4 COMP BYTE 2-3		
32	HPS DDR4 COMP ECC		
33	BLANK		
34	ARDUCAM MIPI		
35	HPS Connector		
36	QSFP1		
37	QSFP2		

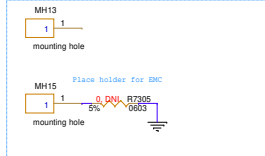
Mounting hole for FPGA



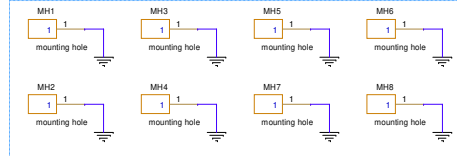
Mounting hole for HPS DC



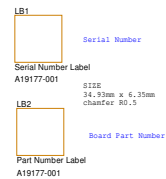
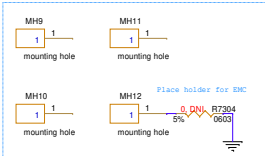
Mounting hole for top VR heatsink



Mounting hole for board

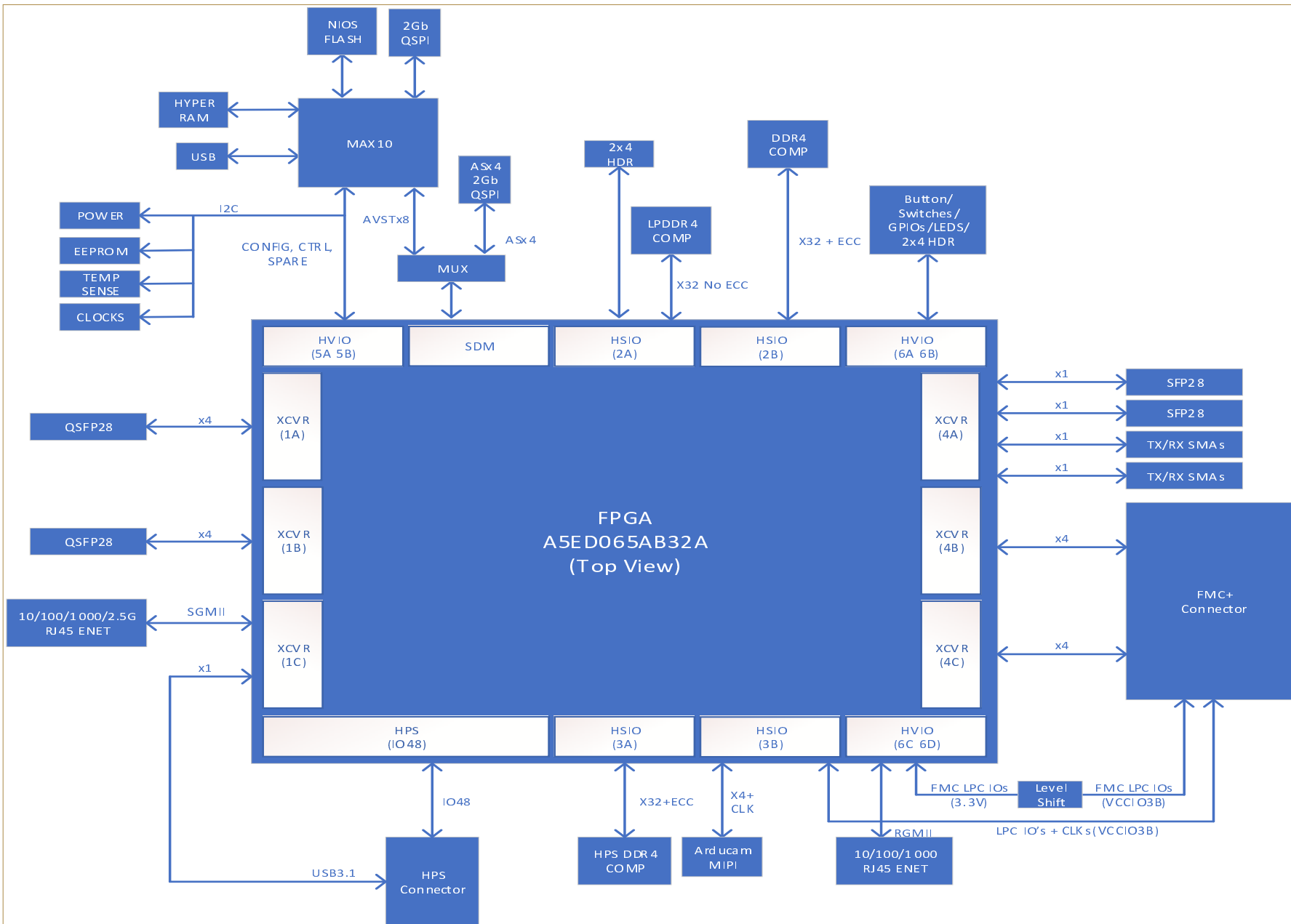


Mounting hole for bottom VR heatsink



altera

Altera Corporation, 101 Innovation Dr., San Jose CA 95134 © 2025 Altera Corporation. All rights reserved.	
File: AGILEX 5 E-SERIES 065A PREMIUM DEV KIT (PRODUCTION)	
Size C	Document Number 190-0330713-A3
Date: Monday, February 09, 2026	Sheet 1 of 67



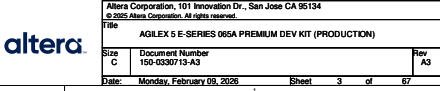
altera

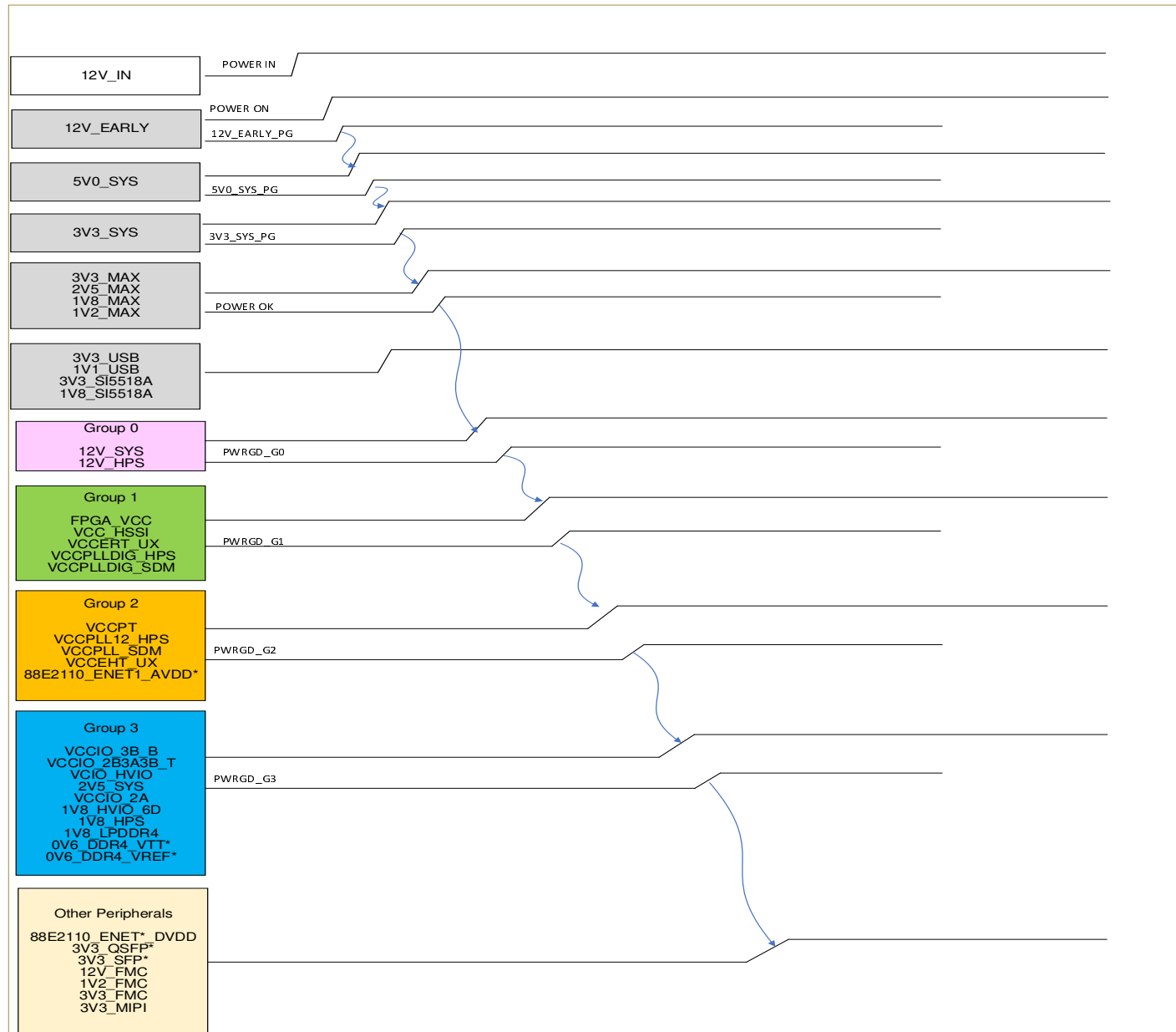
Altera Corporation, 101 Innovation Dr., San Jose CA 95134			
© 2015 Altera Corporation. All rights reserved.			
File	AGILEX 5 E-SERIES 905A PREMIUM DEV KIT (PRODUCTION)		
Size	Document Number	Rev	
C	100-1000715-A3	A3	
Date:	Monday, February 09, 2015	Sheet	2 of 67

```

graph LR
    PWR_ON_SWITCH[PWR_ON SWITCH] --> MAX10((MAX10))
    MAX10 --> Group1((Group1))
    Group1 --> Group2((Group2))
    Group2 --> Group3((Group3))
    Group3 --> Group4((Group4))
    subgraph FPGA
        Group1
        Group2
        Group3
    end

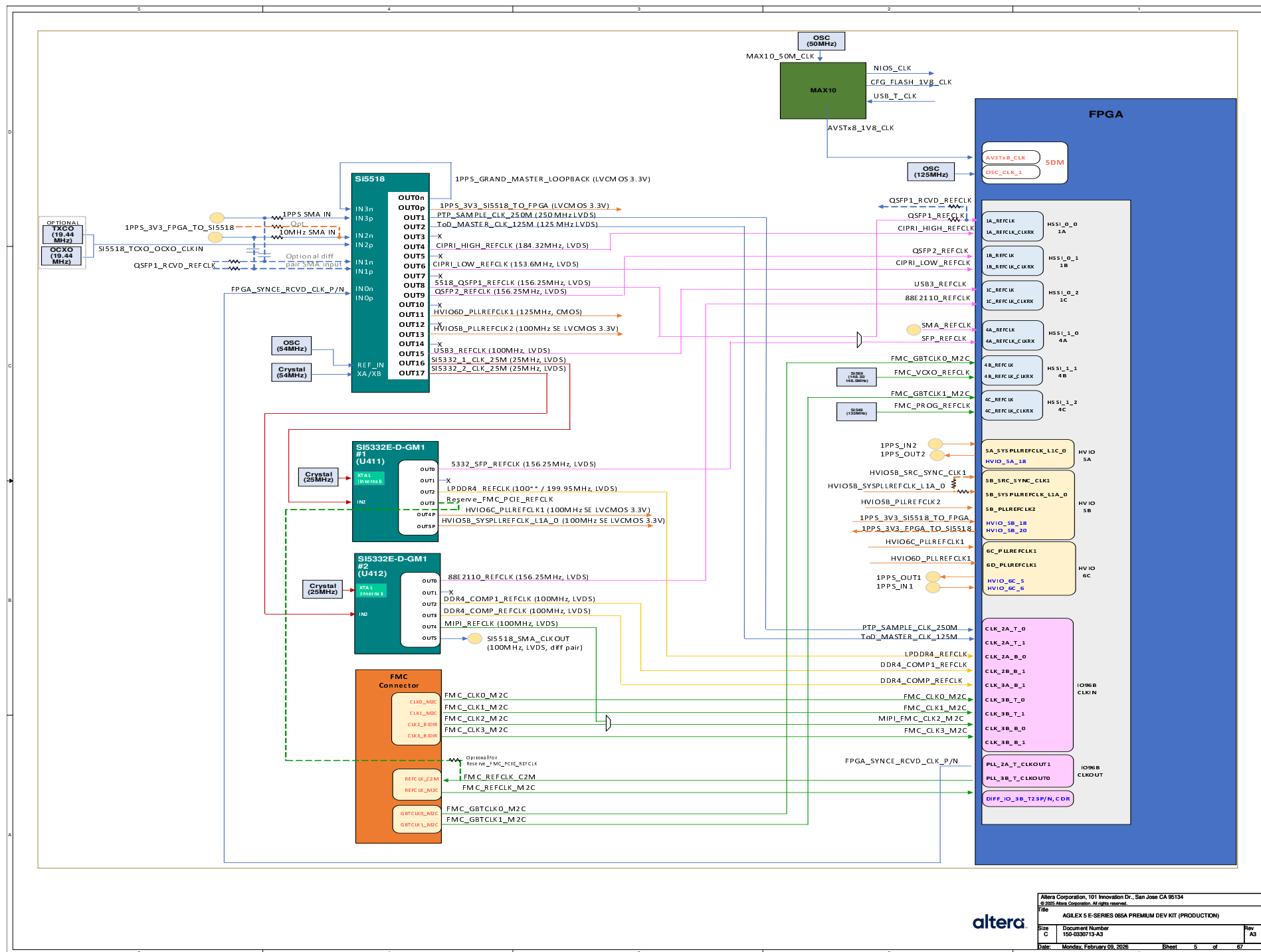
```

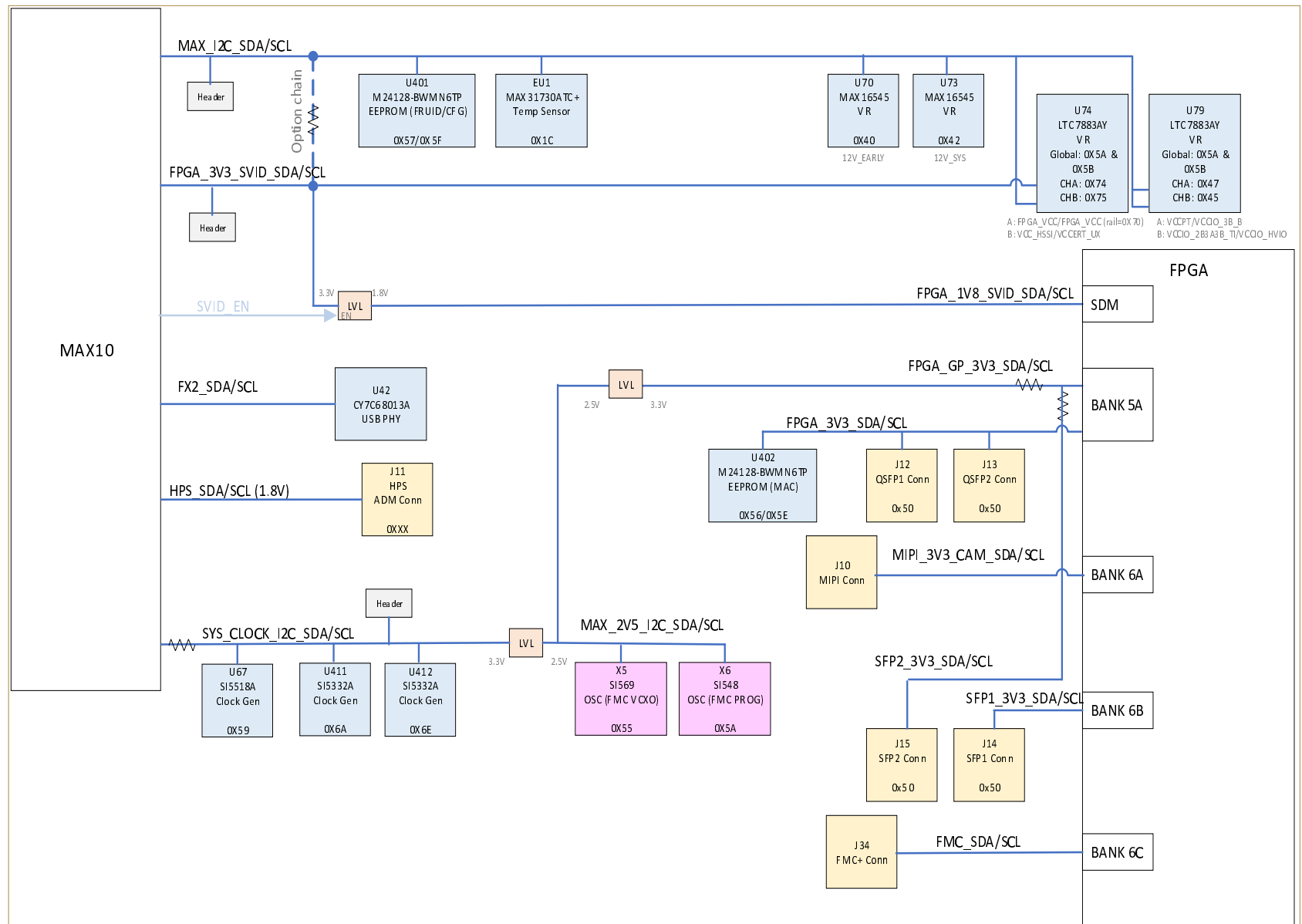




altera

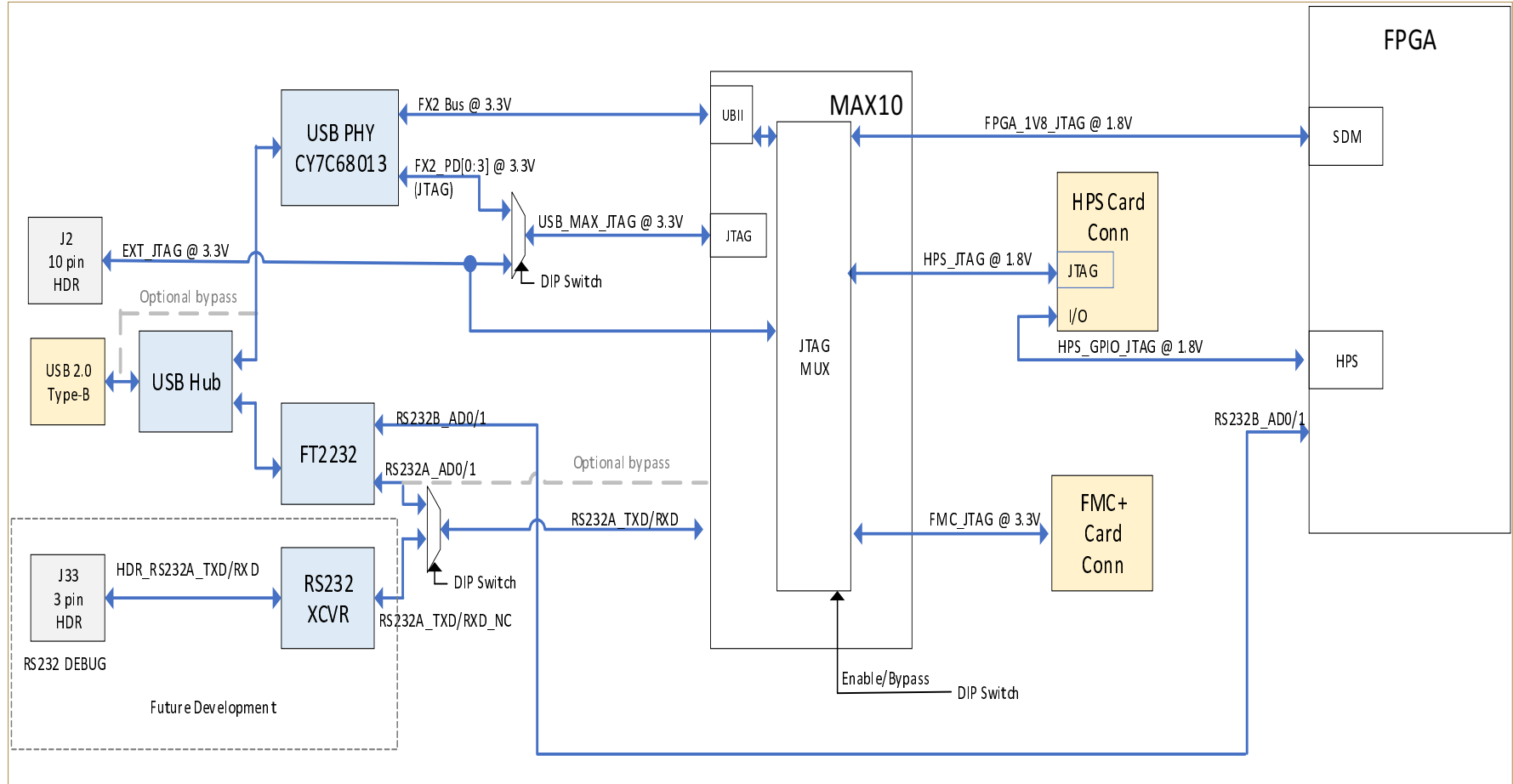
Altera Corporation, 101 Innovation Dr., San Jose CA 95134		
© 2015 Altera Corporation. All rights reserved.		
File: AGILEX 5 E-SERIES 905A PREMIUM DEV KIT (PRODUCTION)		
Size: C	Document Number: 100-1030713-A3	Rev: A3
Date: Monday, February 09, 2016	Sheet: 4 of 67	





altera

Altera Corporation, 161 Innovation Dr., San Jose CA 95134			
© 2015 Altera Corporation. All rights reserved.			
File	AGILEX 5 E-SERIES 905A PREMIUM DEV KIT (PRODUCTION)		
Size	Document Number	Rev	
C	160-0300715-A3	A3	
Date:	Monday, February 09, 2015	Sheet	6 of 67

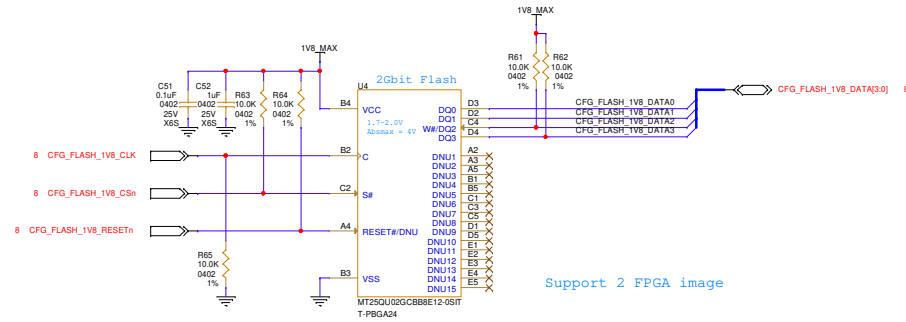


altera

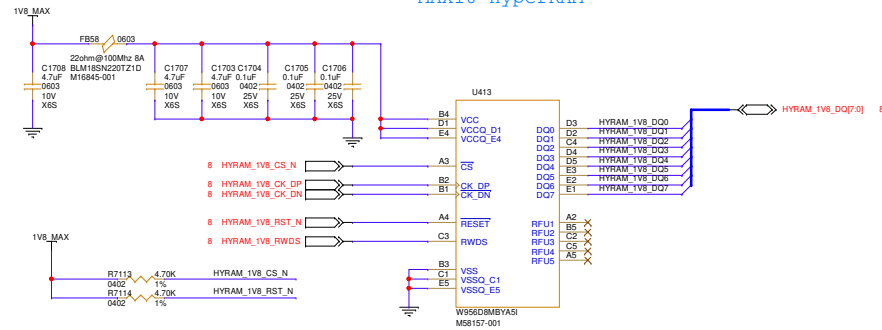
Altera Corporation, 101 Innovation Dr., San Jose CA 95134			
© 2015 Altera Corporation. All rights reserved.			
File: AGILEX 5 E-SERIES 955A PREMIUM DEV KIT (PRODUCTION)			
Size: C	Document Number: 160-1030713-A3	Rev: A3	
Date: Monday, February 09, 2016	Sheet: 7	of 67	



AVST CFG Flash



MAX10 HyperRAM



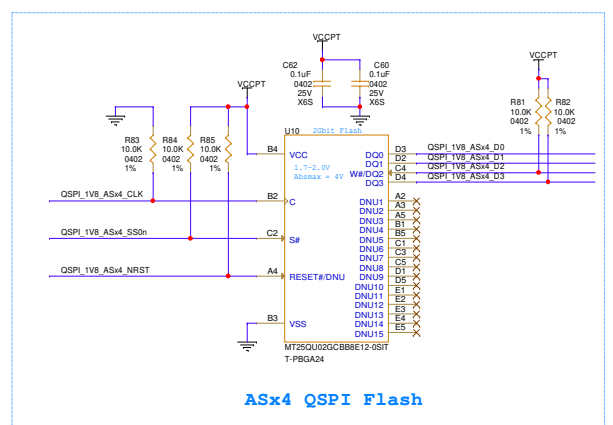
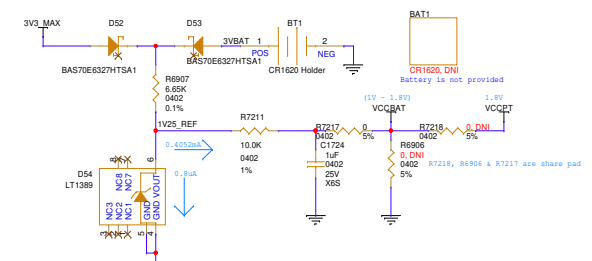
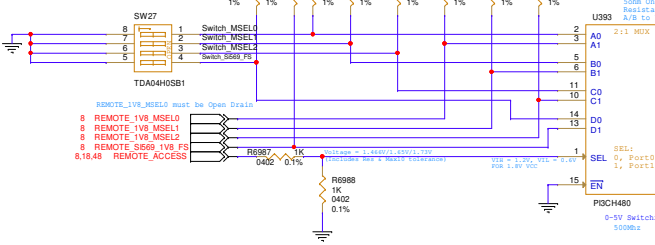
altera

Altera Corporation, 101 Innovation Dr., San Jose CA 95134		
© 2015 Altera Corporation. All rights reserved.		
File: AGILEX 5 E-SERIES 905A PREMIUM DEV KIT (PRODUCTION)		
Size: C	Document Number: 160-0330713-A3	Rev: A3
Date: Monday, February 09, 2016	Sheet: 10 of 67	

SDM CONFIGURATION

SILKSCREEN:
CLOSE = 0, OPEN = 1
1 = MSEL0
2 = MSEL1
3 = MSEL2
4 = S1569_FPGA_SEL

Default Switch Settings:
1-4: Open, 2-3: Close
(Default: A04 Fast, S1569=148,33MHz)



8 AVSTx8_1V8_DATA[7:0]
8 AVSTx8_1V8_READY
8 AVSTx8_1V8_VALID
8 AVSTx8_1V8_CLK

SEL Lo = Port0 <--> Port7 (AVSTx8)
SEL Hi = Port1 <--> Port7 (ASx4, default)

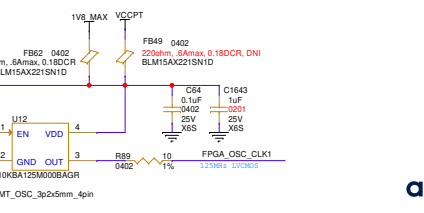
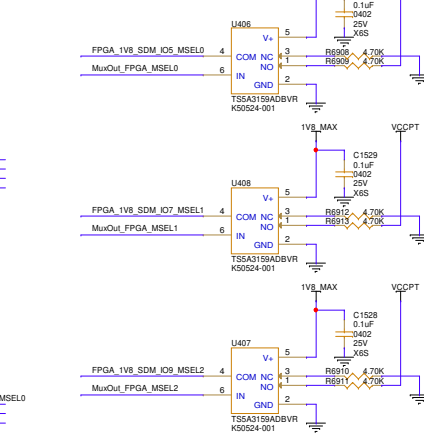
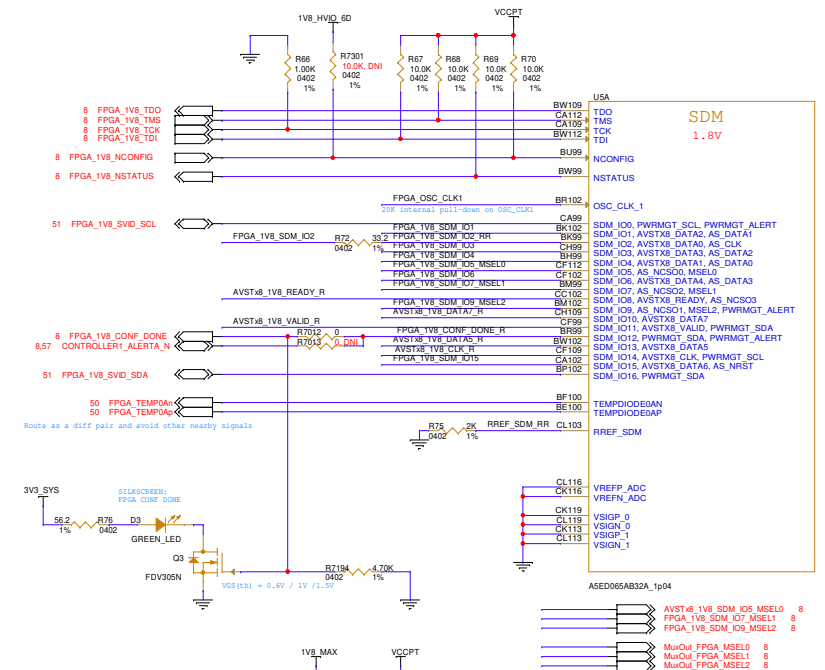
AVSTx8_1V8_READY R7268 0 0402 AVSTx8_1V8_READY_R
AVSTx8_1V8_CLK R7269 0 0402 AVSTx8_1V8_CLK_R
AVSTx8_1V8_DATA7 R7270 0 0402 AVSTx8_1V8_DATA7_R
AVSTx8_1V8_DATA6 R7271 0 0402 AVSTx8_1V8_DATA6_R
AVSTx8_1V8_DATA5 R7272 0 0402 AVSTx8_1V8_DATA5_R
AVSTx8_1V8_DATA4 R7273 0 0402 AVSTx8_1V8_DATA4_R
AVSTx8_1V8_DATA3 R7274 0 0402 AVSTx8_1V8_DATA3_R
AVSTx8_1V8_DATA2 R7275 0 0402 AVSTx8_1V8_DATA2_R
AVSTx8_1V8_DATA1 R7276 0 0402 AVSTx8_1V8_DATA1_R
AVSTx8_1V8_DATA0 R7277 0 0402 AVSTx8_1V8_DATA0_R

QSPI_1V8_ASx4_D1 R964 0 0402 ASx4_1V8_SDM_I01
QSPI_1V8_ASx4_CLK R965 0 0402 ASx4_1V8_SDM_I02
QSPI_1V8_ASx4_D2 R966 0 0402 ASx4_1V8_SDM_I03
QSPI_1V8_ASx4_D0 R967 0 0402 ASx4_1V8_SDM_I04
QSPI_1V8_ASx4_SS0n R968 0 0402 ASx4_1V8_SDM_I05
QSPI_1V8_ASx4_D3 R969 0 0402 ASx4_1V8_SDM_I06
QSPI_1V8_ASx4_NRST R970 0 0402 ASx4_1V8_SDM_I015

AVSTx8_1V8_DATA2 R80 10.0K 0402 1%
ASx4_1V8_SDM_I01 U8 211 MUX VCC
AVSTx8_1V8_DATA0 R 5 80
ASx4_1V8_SDM_I02 R 5 80
AVSTx8_1V8_DATA3 R 5 80
ASx4_1V8_SDM_I03 R 5 80
AVSTx8_1V8_DATA1 R 5 80
ASx4_1V8_SDM_I04 R 5 80
AVSTx8_1V8_DATA4 R 5 80
ASx4_1V8_SDM_I05 R 5 80
AVSTx8_1V8_DATA5 R 5 80
ASx4_1V8_SDM_I06 R 5 80
AVSTx8_1V8_DATA6 R 5 80
ASx4_1V8_SDM_I015 R 5 80

AVSTx8_1V8_SDM_I05 MSEL0 U11 211 MUX VCC
ASx4_1V8_SDM_I05 R 3 80
AVSTx8_1V8_DATA4 R 5 80
ASx4_1V8_SDM_I06 R 5 80
AVSTx8_1V8_DATA6 R 5 80
ASx4_1V8_SDM_I015 R 5 80

AVSTx8_1V8_SDM_I05 MSEL0 U11 211 MUX VCC
ASx4_1V8_SDM_I05 R 3 80
AVSTx8_1V8_DATA4 R 5 80
ASx4_1V8_SDM_I06 R 5 80
AVSTx8_1V8_DATA6 R 5 80
ASx4_1V8_SDM_I015 R 5 80



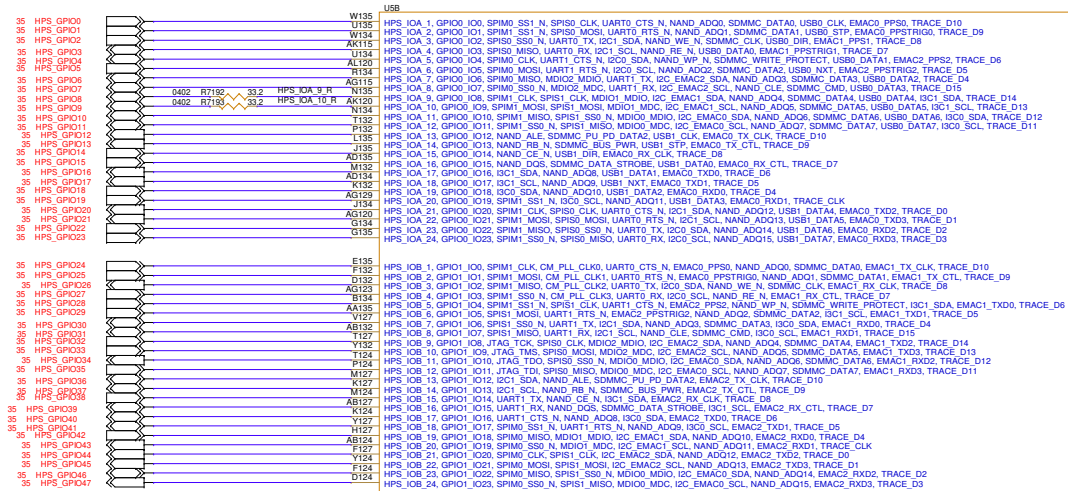
Pins	MSEL	CFG_MUX_SEL = 0	CFG_MUX_SEL = 1
SDM_I00	MSEL	Avalon-ST x8	AS (normal mode)
SDM_I01	MSEL	PWR_SCL	PWR_SCL
SDM_I02	MSEL	AVSTx8_DATA2	AS_DATA1
SDM_I03	MSEL	AVSTx8_DATA0	AS_CLK
SDM_I04	MSEL	AVSTx8_DATA3	AS_DATA2
SDM_I05	MSEL	AVSTx8_DATA1	AS_DATA0
SDM_I06	MSEL	AVSTx8_DATA4	AS_nCS00
SDM_I07	MSEL	AVSTx8_DATA5	AS_nCS01
SDM_I08	MSEL	AVSTx8_DATA6	AS_nCS02
SDM_I09	MSEL	AVSTx8_DATA7	AS_nCS03
SDM_I010	MSEL	AVSTx8_VALID	HPS_COLD_nRESET
SDM_I011	MSEL	AVSTx8_READY	HPS_COLD_nRESET
SDM_I012	MSEL	CONF_DONE	CONF_DONE
SDM_I013	MSEL	AVSTx8_DATA5	CONF_DONE
SDM_I014	MSEL	AVSTx8_CLK	CONF_DONE
SDM_I015	MSEL	AVSTx8_DATA6	AS_nRST
SDM_I016	MSEL	PWR_SDA	PWR_SDA

CFG_MUX_SEL	MSEL0	MSEL1	MSEL2
Default	0	0	1
ASx4 NORM	0	1	1
AVSTx8	1	1	0
JTAG	1	1	1

Note: MSEL0 is shared with QSPI_S0n which has a pull-up. MSEL0 should make MSEL0 open drain.



HPS BANK IO48



ASED0605AB32A_1p048

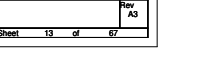
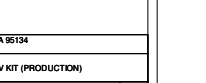
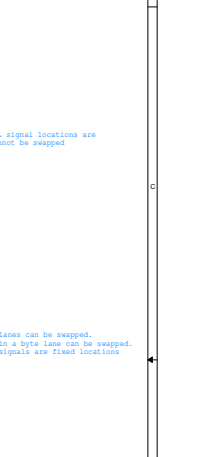
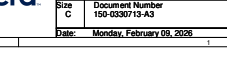
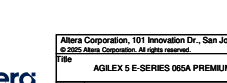
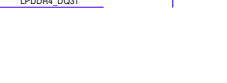
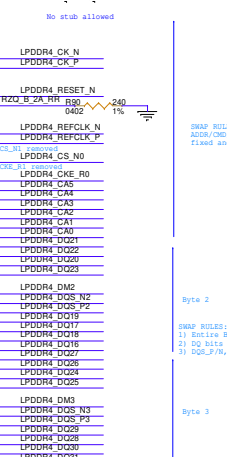
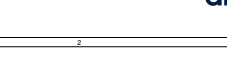
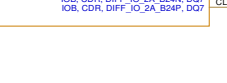
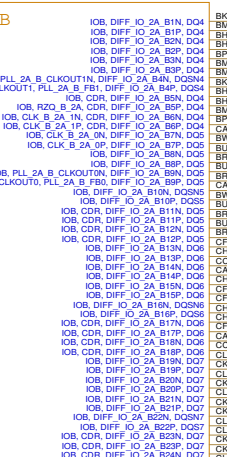
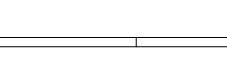
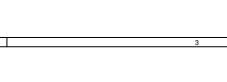
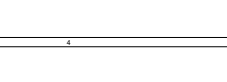
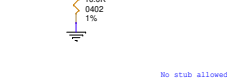
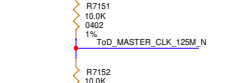
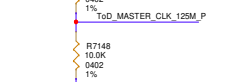
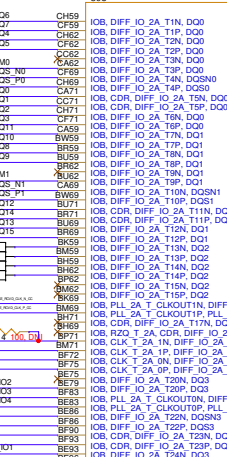
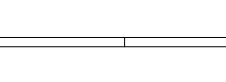
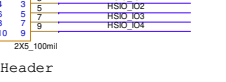
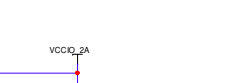
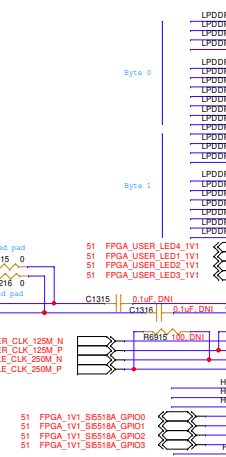
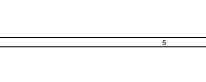
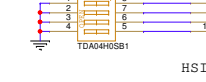
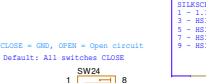
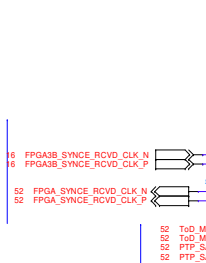
HSIO BANKS 2AL AND 2AR

Please refer to Agilent EMIF User Guide for pin assignment rules.

- SWAP RULES:
- 1) Entire Byte Lanes can be swapped.
 - 2) DQ bits within a byte lane can be swapped.
 - 3) DQ-P/N, DM signals are fixed locations

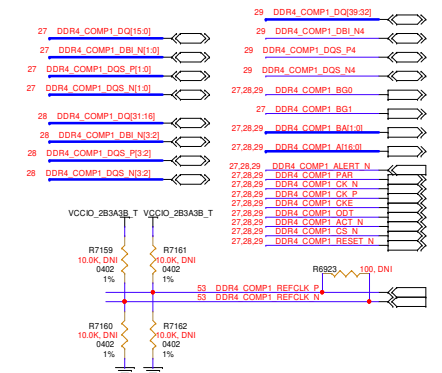
SWIZZLE RULES:

BYTE_SWIZZLE_CM[0:3] = 2, 0, 3, 1, 0, 1, 3, 2;
 PTH_SWIZZLE_CM[0:3] = 15, 13, 14, 12, 9, 8, 10, 11;
 PTH_SWIZZLE_CM[0:3] = 18, 17, 16, 15, 20, 22, 21;
 PTH_SWIZZLE_CM[0:3] = 31, 30, 29, 28, 25, 24, 26, 27;



HSIO BANKS 2BL AND 2BR

Please refer to Agilex EMIF User Guide for pin assinging rules.



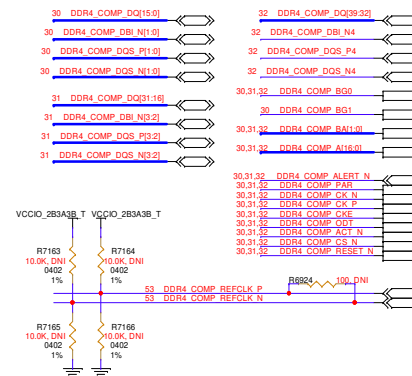
Place near R2Q pin

SWAP RULES:
A058/ODT/CTRS signal locations are fixed and cannot be swapped

altera

Altera Corporation, 161 Innovation Dr., San Jose CA 95134			
© 2016 Altera Corporation. All rights reserved.			
File	AGILEX 5 E-SERIES 905A PREMIUM DEV KIT (PRODUCTION)		
Size	Document Number	160-0330713-A3	Rev A3
Date:	Monday, February 09, 2016	Sheet	14 of 67

Please refer to [Agilex EMIE User Guide](#) for pin assignment rules.



SWAP RULES:
ADDR/CMD/CTRL signal locations are
fixed and cannot be swapped

Steve G

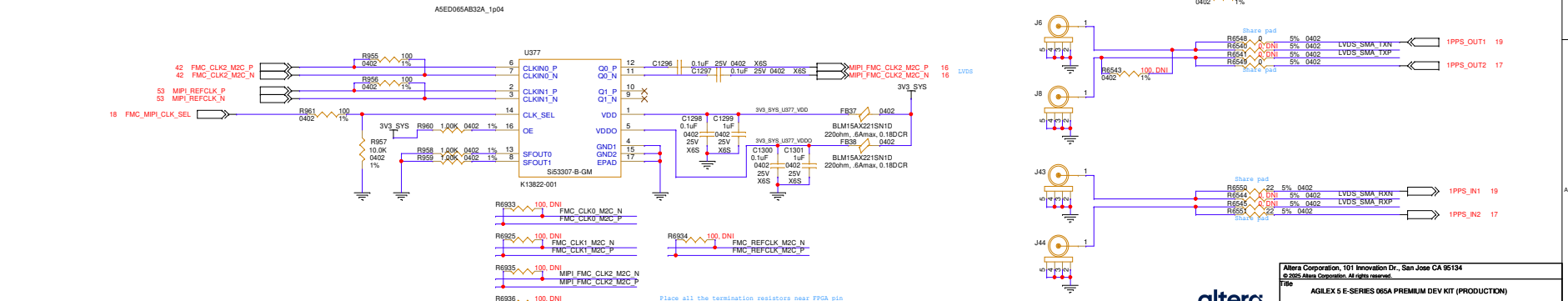
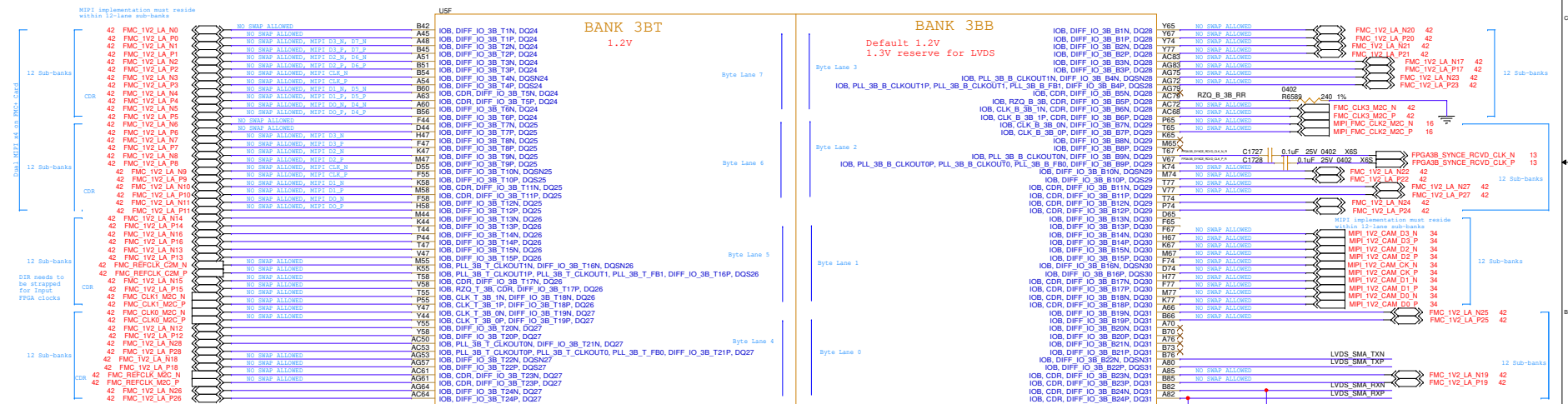
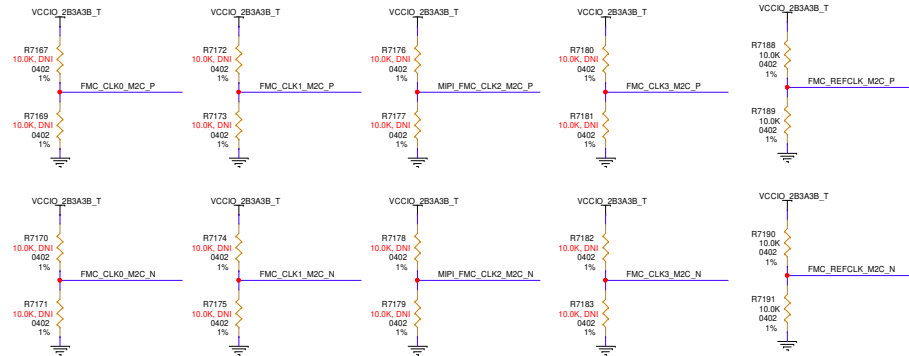
Pin placement restriction for MIPI & LVDS mixing with LVCMOS:

Restricted Pin Used As LVCMOS	
IO Lane Used for MIPI	Width
IO Lane 1	1 data + 1 clock
	2 data + 1 clock
	4 data + 1 clock
IO Lane 6	1 data + 1 clock
	2 data + 1 clock
	4 data + 1 clock
IO Lane 7	1 data + 1 clock
	2 data + 1 clock
	4 data + 1 clock
IO Lane 6 and 7	8 data + 1 clock

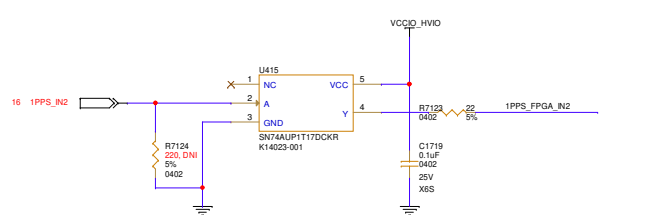
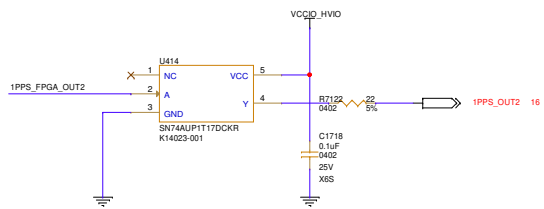
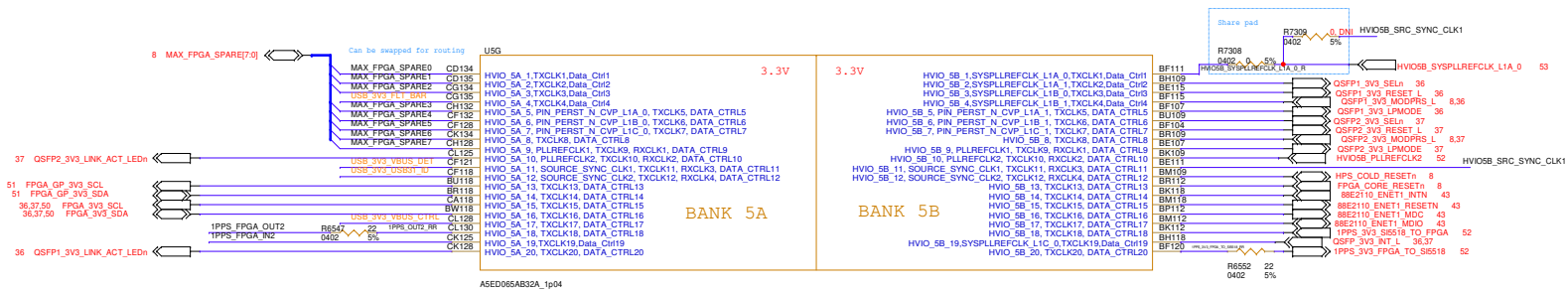
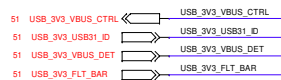
Restricted Pin Used As LVCMOS	
IO Lane Used for LVDS	Function
IO Lane 0	LVDS_SMA_RX
IO Lane 2	FPGA3B_SYNC_RCVD_CLK
IO Lane 3	MIPI_FMC_CLK2_M2C
IO Lane 4	FMC_CLK3_M2C
IO Lane 5	FMC_REFCLK_M2C
IO Lane 6	FMC_CLK0_M2C
IO Lane 7	FMC_REFCLK_C2M

NOTES: Please check the detail restriction guide from IO Pinout Guideline Document if any of the FMC_V12_LA_* to be used as LVDS.

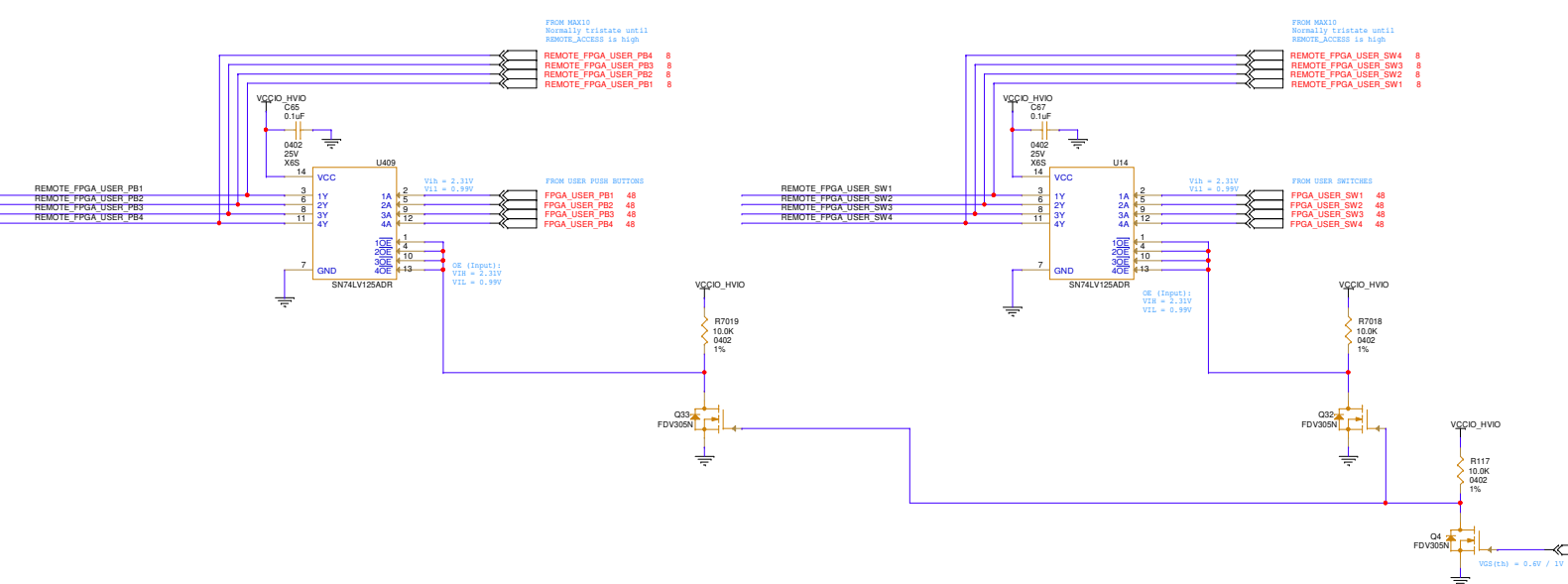
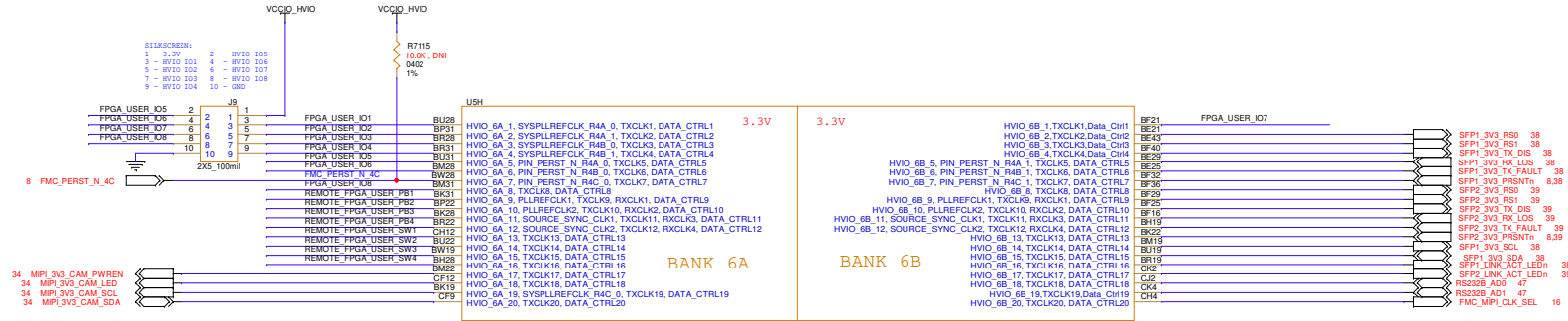
HSIO BANKS 3BL AND 3BR



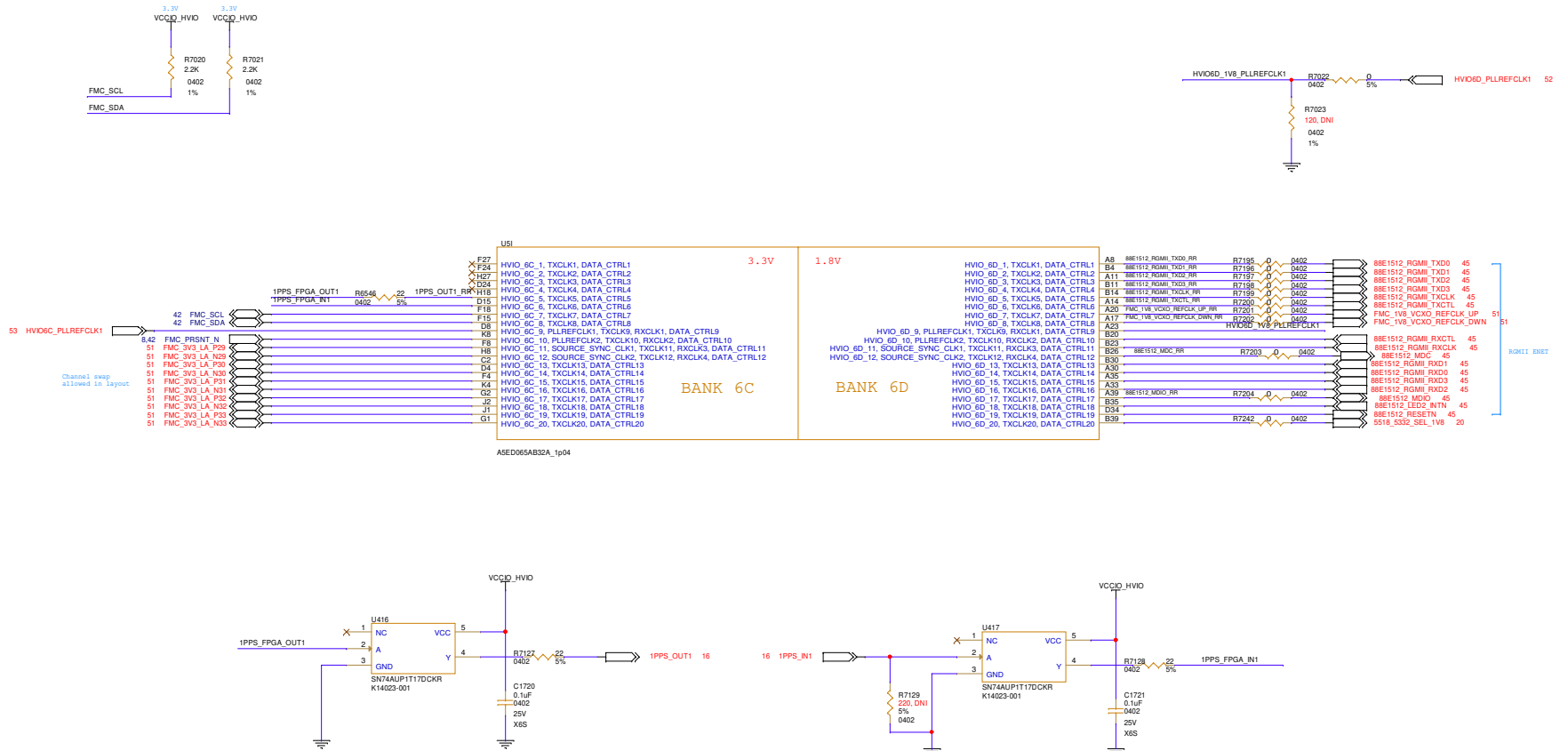
HVIO BANKS 5A AND 5B



HVIO BANKS 6A AND 6B



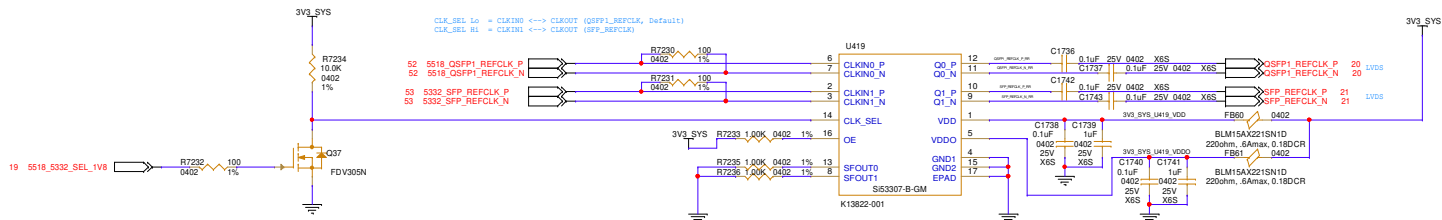
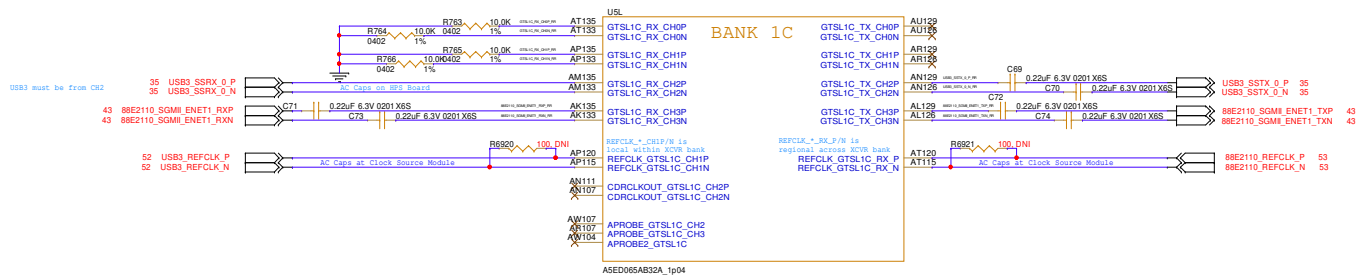
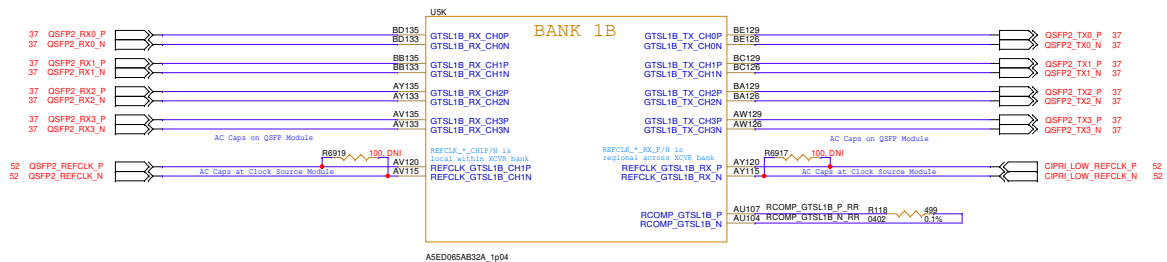
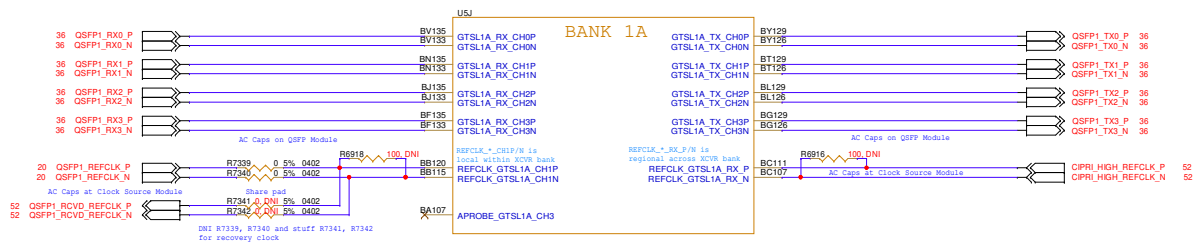
HVIO BANKS 6C AND 6D



altera

Altera Corporation, 161 Innovation Dr., San Jose CA 95134			
© 2005 Altera Corporation. All rights reserved.			
File: AGILEX 5 E-SERIES 905A PREMIUM DEV KIT (PRODUCTION)			
Size: C	Document Number: 160-0300715-A3	Rev: A3	
Date: Monday, February 09, 2006	Sheet: 19	of 67	

XCVR BANKS 1A, 1B, 1C



DNU & NC PINS

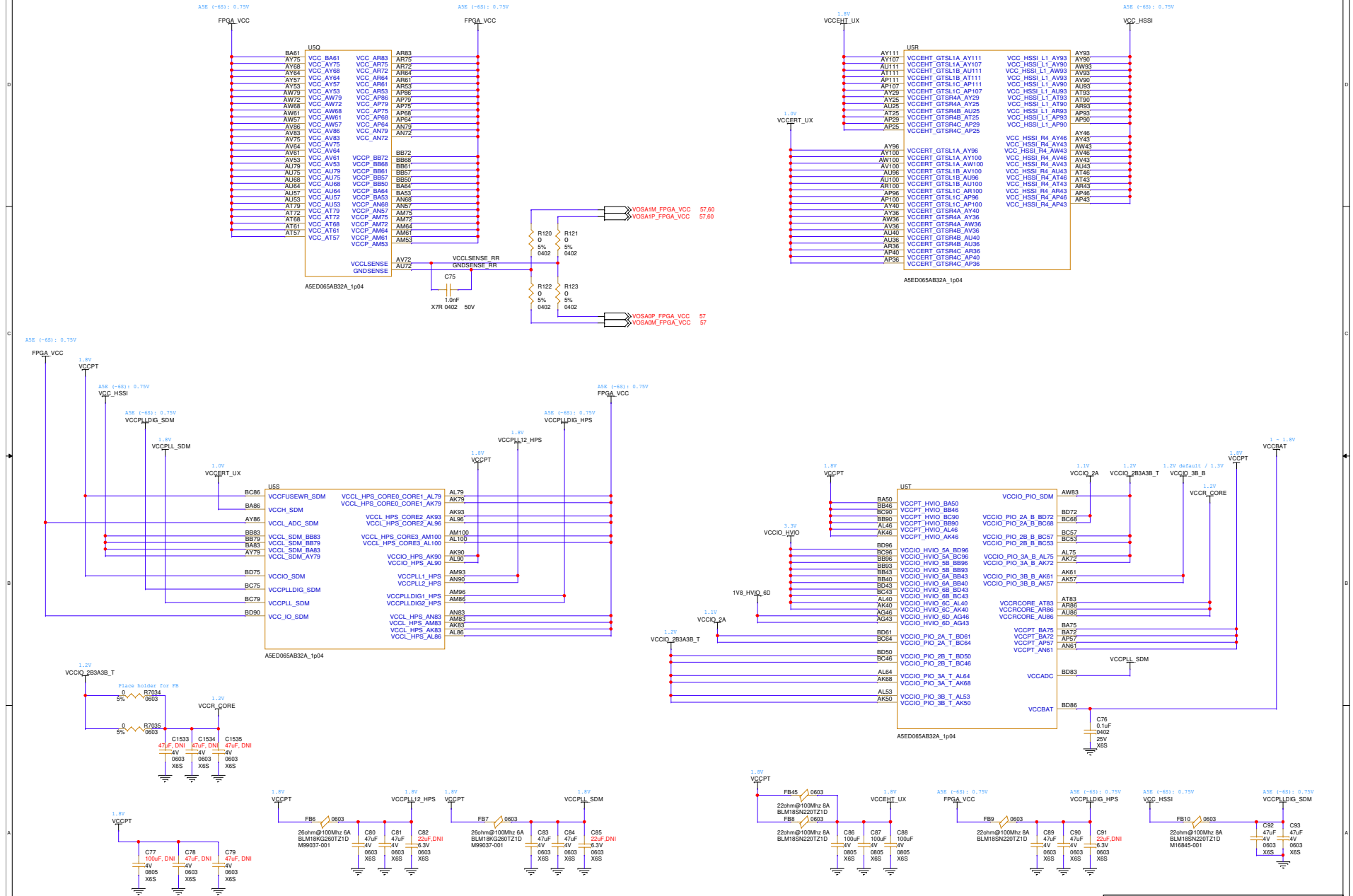
USP	
CL101	DNU CL101
CL101	DNU CK101
CL103	DNU CK103
CL106	DNU CL106
AM120	DNU AM120
AL115	DNU AL115
BD64	DNU BD64
BD53	DNU BD53
AL68	DNU AL68
AL57	DNU AL57
BD120	DNU BD120
RC40	DNU RC40
AL43	DNU AL43
BA104	DNU BA104
BA32	DNU BA32
AR104	DNU AR104
AR32	DNU AR32
BD100	DNU BD100
BD36	DNU BD36
BD100	DNU BD100
BC36	DNU BC36
AM36	DNU AM36

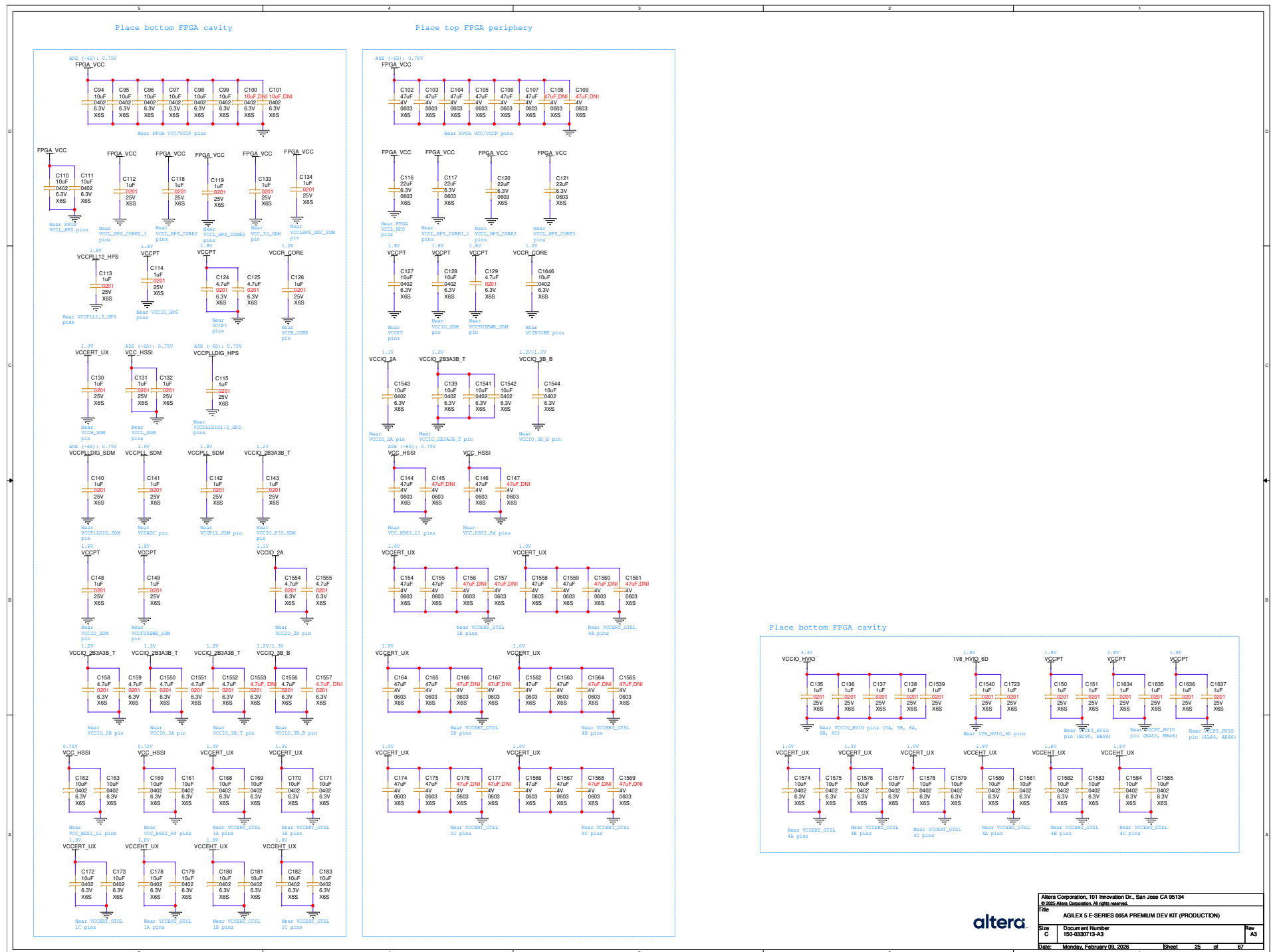
ASED065AB32A_1p04

USX	
Y8	NC Y8
Y37	NC Y4
Y34	NC Y37
Y27	NC Y34
Y24	NC Y27
Y18	NC Y24
Y15	NC Y18
W2	NC Y15
W1	NC W2
W8	NC W1
V37	NC V8
V27	NC V37
V18	NC V27
U2	NC V18
U1	NC U2
T8	NC U1
T4	NC T8
T37	NC T4
T34	NC T37
T27	NC T34
T24	NC T27
T18	NC T24
T15	NC T18
R2	NC T15
R4	NC R2
P4	NC P4
P24	NC P4
P15	NC P24
N2	NC P15
N1	NC N2
M8	NC N1
M4	NC M8
M37	NC M4
M34	NC M37
M27	NC M34
M24	NC M27
M18	NC M24
M15	NC M18
L1	NC M15
K37	NC L1
K34	NC K37
K27	NC K34
K24	NC K27
K18	NC K24
K15	NC K18
H37	NC K15
F37	NC H37
F34	NC F37
BD16	NC F34
AY50	NC BD16
AY50	NC AY50

ASED065AB32A_1p04

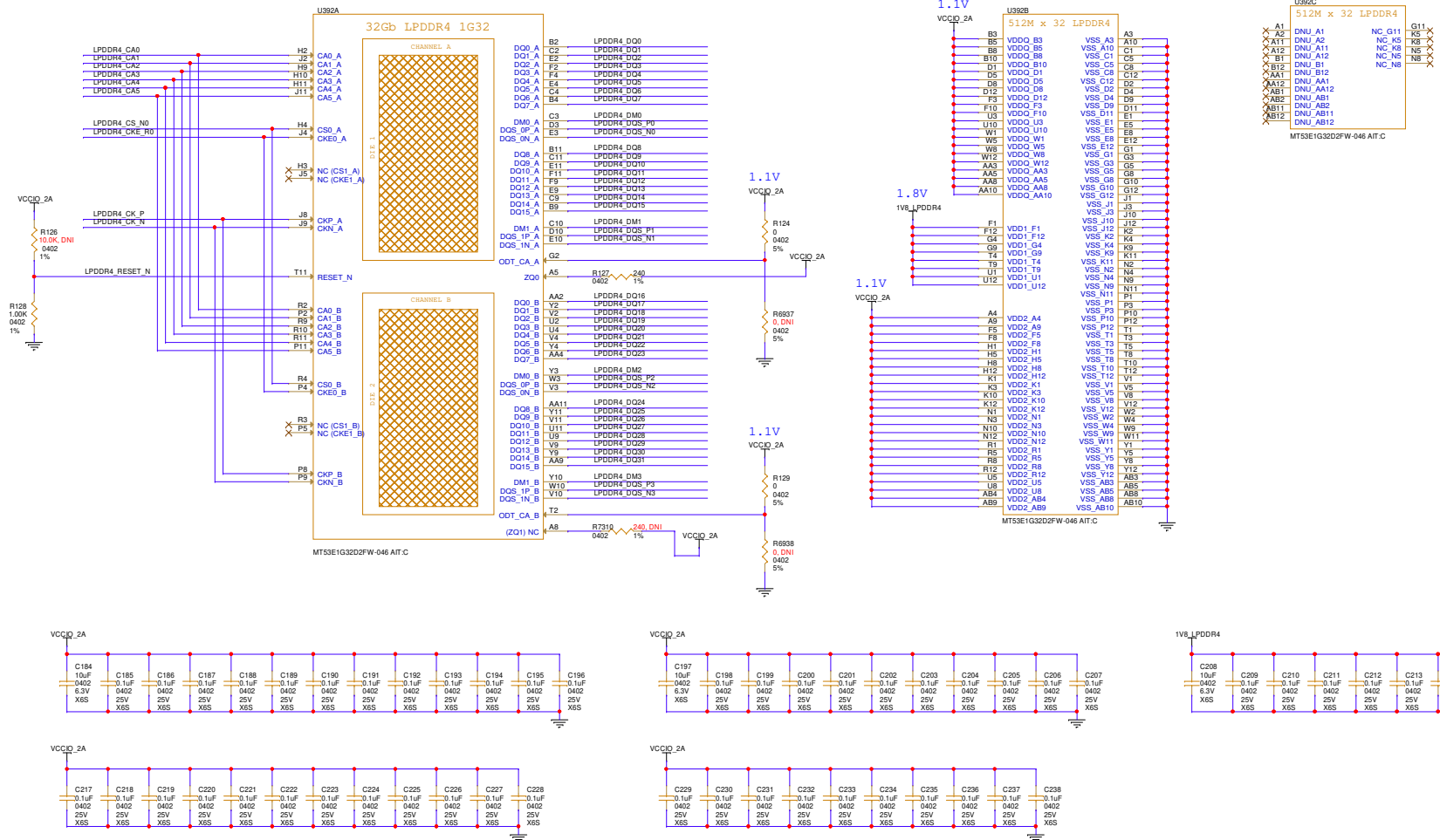
FPGA POWER



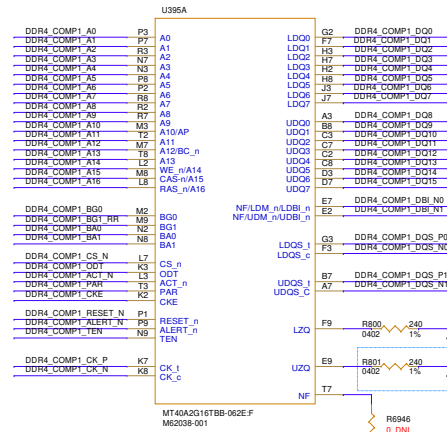
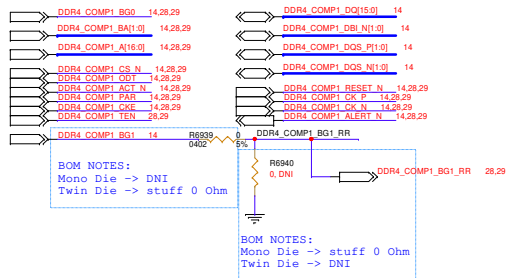


32-BIT LPDDR4

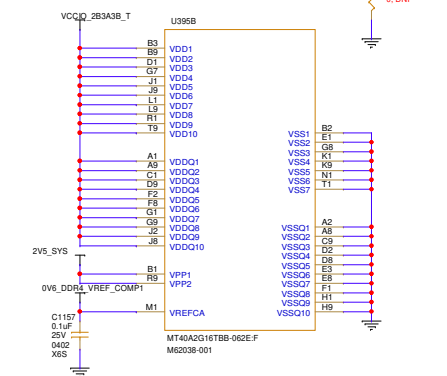
LPDDR4_DQ[31:0] 13
 LPDDR4_DM[3:0] 13
 LPDDR4_DQS_P[3:0] 13
 LPDDR4_DQS_N[3:0] 13
 LPDDR4_CA[5:0] 13
 LPDDR4_CK_P 13
 LPDDR4_CK_N 13
 LPDDR4_RESET_N 13
 LPDDR4_CS_N0 13
 LPDDR4_CKE_R0 13



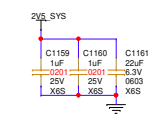
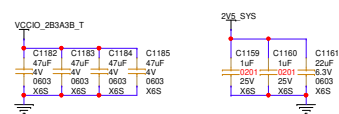
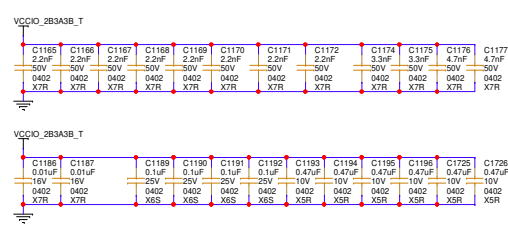
8GB DDR4 COMP1 BYTE 0-1



MT40A2G16 is TwinDie build up from 2x MT40A2G8 (16Gb x8)
For Mono Die device, Pin E9 = GND; M9 = GND
For Twin Die device, Pin E9 = U2Q; M9 = BGL.



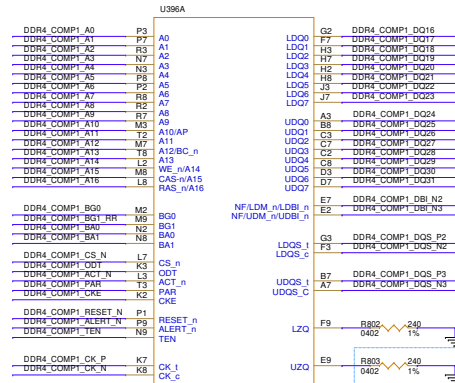
BOM NOTES:
Mono Die -> stuff 0 Ohm
Twin Die -> stuff 240 Ohm



8GB DDR4 COMP1 BYTE 2-3

DDR4 COMP1_B00	14,27,29
DDR4 COMP1_BA1[0]	14,27,29
DDR4 COMP1_A[16:0]	14,27,29
DDR4 COMP1_CS_N	14,27,29
DDR4 COMP1_ODT	14,27,29
DDR4 COMP1_ACT_N	14,27,29
DDR4 COMP1_PAR	14,27,29
DDR4 COMP1_CKE	14,27,29
DDR4 COMP1_TEN	27,29
DDR4 COMP1_BG1_BB	27,29

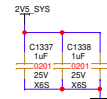
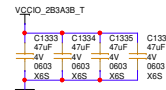
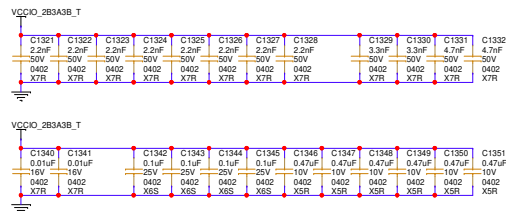
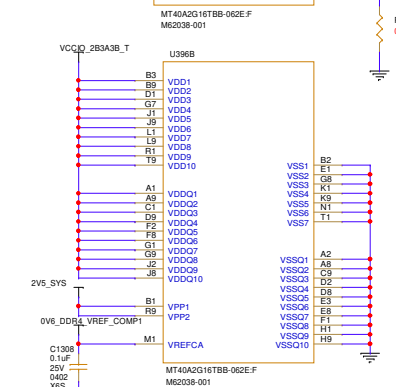
DDR4 COMP1_DQ[31:16]	14
DDR4 COMP1_DBI_N[3:2]	14
DDR4 COMP1_DQS_P[3:2]	14
DDR4 COMP1_DQS_N[3:2]	14
DDR4 COMP1_RESET_N	14,27,29
DDR4 COMP1_CK_P	14,27,29
DDR4 COMP1_CK_N	14,27,29
DDR4 COMP1_ALERT_N	14,27,29



MT40A2G16 is TwinDie build up from 2x MT40A2G8 (16Gb x8)

For Mono Die device, Pin E9 = GND; M9 = GND
For Twin Die device, Pin E9 = U2Q; M9 = BG1.

BOM NOTES:
Mono Die -> stuff 0 Ohm
Twin Die -> stuff 240 Ohm



altera

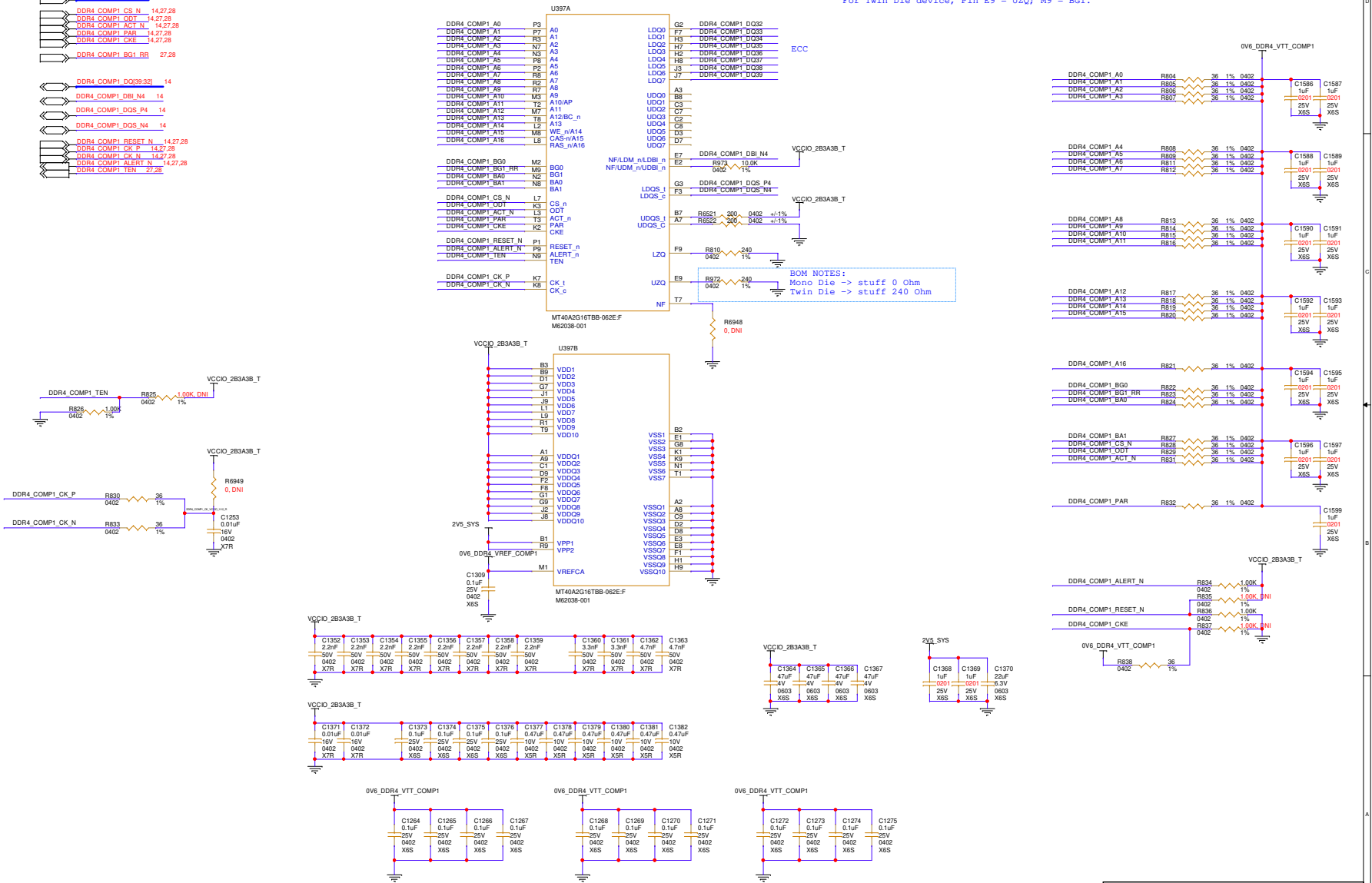
Altera Corporation, 161 Innovation Dr., San Jose CA 95134			
© 2005 Altera Corporation. All rights reserved.			
File	AGILEX 5 E-SERIES 905A PREMIUM DEV KIT (PRODUCTION)		
Size	Document Number	Rev	
C	160-0330713-A3	A3	
Date:	Monday, February 09, 2006	Sheet	28 of 67

8GB DDR4 COMP1 BYTE ECC

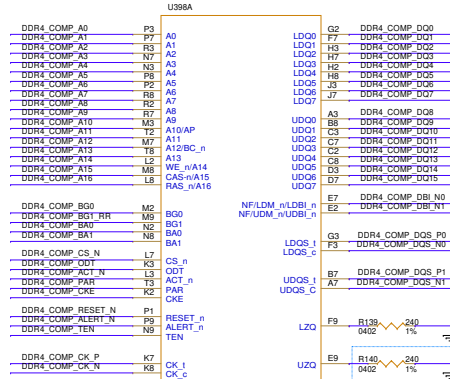
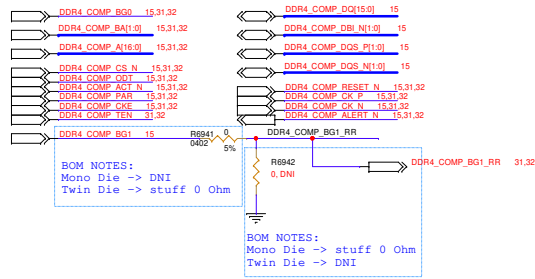
DDR4_COMP1_B00 14,27,28
 DDR4_COMP1_BA[1:0] 14,27,28
 DDR4_COMP1_A[16:0] 14,27,28
 DDR4_COMP1_CS_N 14,27,28
 DDR4_COMP1_ODT 14,27,28
 DDR4_COMP1_ACT_N 14,27,28
 DDR4_COMP1_PAR 14,27,28
 DDR4_COMP1_CKE 14,27,28
 DDR4_COMP1_B01_RR 27,28
 DDR4_COMP1_DQS_B32 14
 DDR4_COMP1_DBI_N4 14
 DDR4_COMP1_DQS_P4 14
 DDR4_COMP1_DQS_N4 14
 DDR4_COMP1_RESET_N 14,27,28
 DDR4_COMP1_CK_P 14,27,28
 DDR4_COMP1_CK_N 14,27,28
 DDR4_COMP1_ALERT_N 14,27,28
 DDR4_COMP1_TEN 27,28

MT40A2G16 is TwinDie build up from 2x MT40A2G8 (16Gb x8)

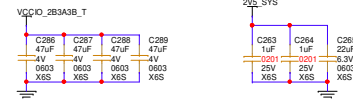
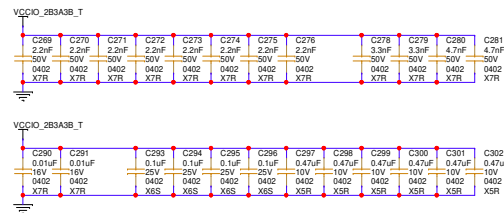
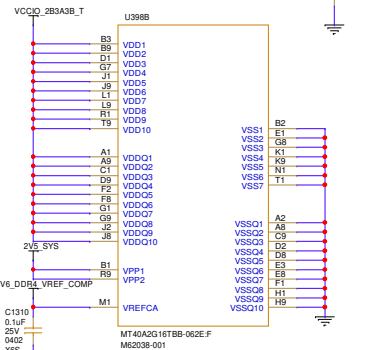
For Mono Die device, Pin E9 = GND; M9 = GND
 For Twin Die device, Pin E9 = UZQ; M9 = BG1.



HPS 8GB DDR4 COMPONENT BYTE 0-1



MT40A2G16 is TwinDie build up from 2x MT40A2G8 (16Gb x8)
For Mono Die device, Pin E9 = GND; M9 = GND
For Twin Die device, Pin E9 = UZQ; M9 = BG1.

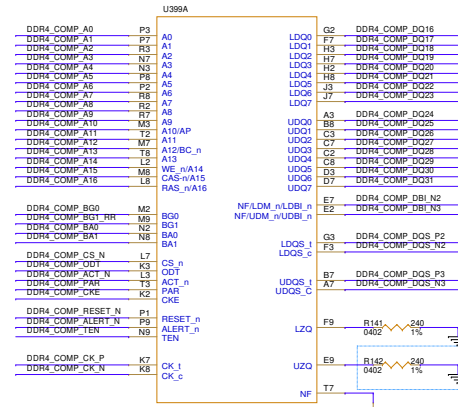


altera

HPS 8GB DDR4 COMPONENT BYTE 2-3

DDR4_COMP_BG0	15,30,32
DDR4_COMP_BA[1:0]	15,30,32
DDR4_COMP_A[16:0]	15,30,32
DDR4_COMP_CS_N	15,30,32
DDR4_COMP_OBT	15,30,32
DDR4_COMP_ACT_N	15,30,32
DDR4_COMP_PAR	15,30,32
DDR4_COMP_CKE	15,30,32
DDR4_COMP_TEN	30,32
DDR4_COMP_BG1_RR	30,32

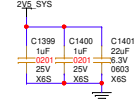
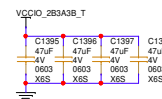
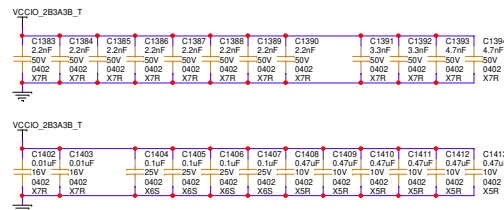
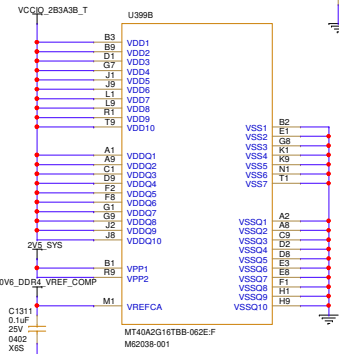
DDR4_COMP_DQ[31:16]	15
DDR4_COMP_DBI_N[3:2]	15
DDR4_COMP_DQS_P[3:2]	15
DDR4_COMP_DQS_N[3:2]	15
DDR4_COMP_RESET_N	15,30,32
DDR4_COMP_CK_P	15,30,32
DDR4_COMP_CK_N	15,30,32
DDR4_COMP_ALERT_N	15,30,32



MT40A2G16 is TwinDie build up from 2x MT40A2G8 (16Gb x8)

For Mono Die device, Pin E9 = GND; M9 = GND
For Twin Die device, Pin E9 = U2Q; M9 = BG1.

BOM NOTES:
Mono Die -> stuff 0 Ohm
Twin Die -> stuff 240 Ohm



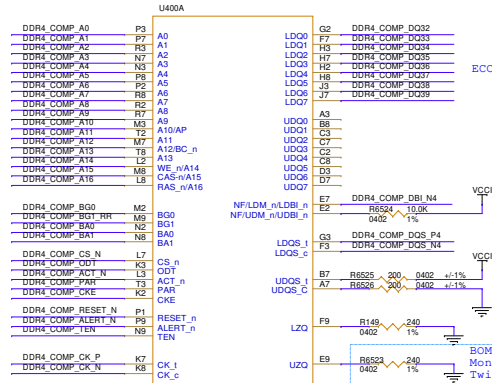
altera

Altera Corporation, 161 Innovation Dr., San Jose CA 95134			
© 2015 Altera Corporation. All rights reserved.			
File	AGILEX 5 E-SERIES 905A PREMIUM DEV KIT (PRODUCTION)		
Size	Document Number	Rev	
C	160-0300715-A3	A3	
Date:	Monday, February 09, 2016	Sheet	31 of 67

HPS 8GB DDR4 COMPONENT BYTE ECC

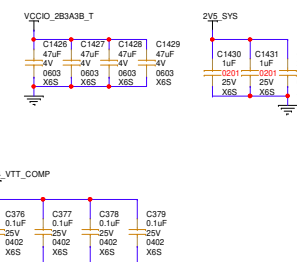
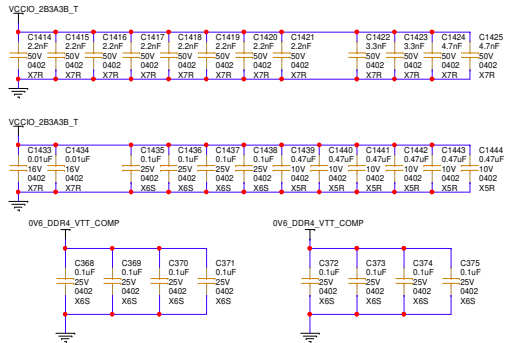
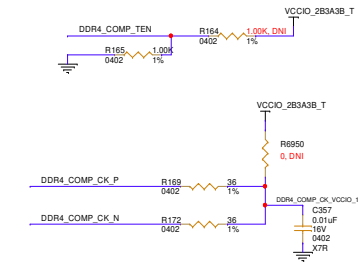
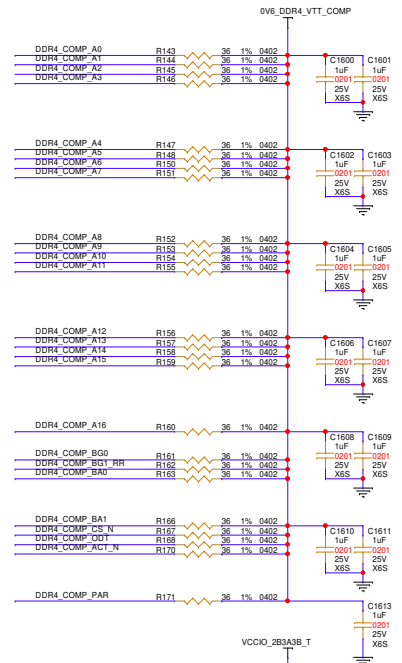
DDR4_COMP_BG0 15,30,31
DDR4_COMP_BA[1:0] 15,30,31
DDR4_COMP_A[16:0] 15,30,31
DDR4_COMP_CS_N 15,30,31
DDR4_COMP_C0[7] 15,30,31
DDR4_COMP_ACT_N 15,30,31
DDR4_COMP_PAB 15,30,31
DDR4_COMP_CKE 15,30,31
DDR4_COMP_BG1_RR 30,31

DDR4_COMP_DQ[9:3] 15
DDR4_COMP_DBI_N4 15
DDR4_COMP_DQS_P4 15
DDR4_COMP_DQS_N4 15
DDR4_COMP_RESET_N 15,30,31
DDR4_COMP_CK_P 15,30,31
DDR4_COMP_CK_N 15,30,31
DDR4_COMP_ALERT_N 15,30,31
DDR4_COMP_TEN 30,31



BOM NOTES:
Mono Die -> stuff 0 Ohm
Twin Die -> stuff 240 Ohm

MT40A2G16 is TwinDie build up from 2x MT40A2G8 (16Gb x8)
For Mono Die device, Pin E9 = GND; M9 = GND
For Twin Die device, Pin E9 = U2Q; M9 = BG1.



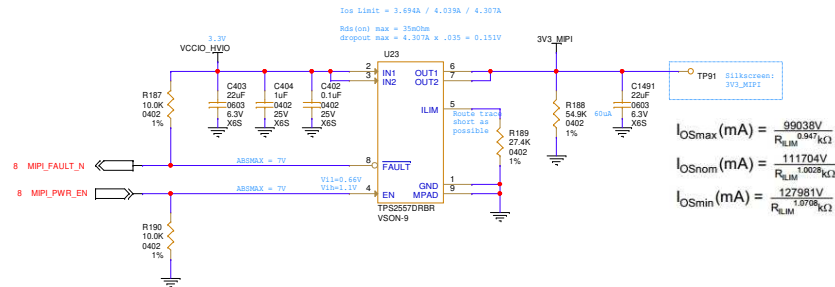
altera

BLANK

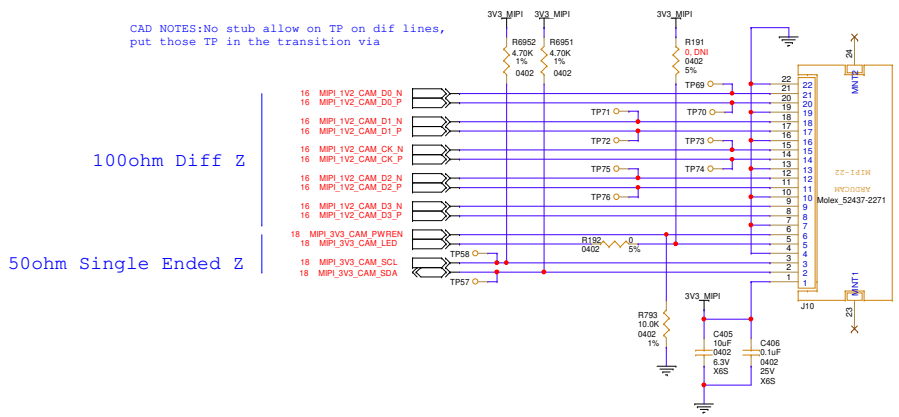


Altera Corporation, 101 Innovation Dr., San Jose CA 95134
© 2015 Altera Corporation. All rights reserved.

Title			
AGILEX 5 E-SERIES 065A PREMIUM DEV KIT (PRODUCTION)			
Size	Document Number		Rev
	150-0330713-A3		
Date: Monday, February 09, 2016		Sheet 33 of	67

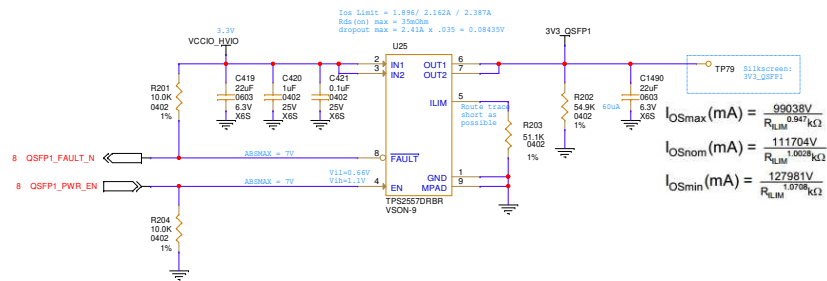


CAD NOTES: No stub allow on TP on dif lines,
put those TP in the transition via



altera

Altera Corporation, 161 Innovation Dr., San Jose CA 95134			
© 2008 Altera Corporation. All rights reserved.			
File: AGILEX 5 E-SERIES 905A PREMIUM DEV KIT (PRODUCTION)			
Size	Document Number	Rev	
C	160-0330713-A3	A3	
Date:	Monday, February 09, 2008	Sheet	34 of 67

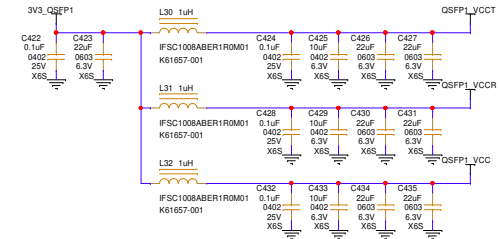


NOTE 1: Bypass Capacitors should be placed as close to the associated 20-pin connector as possible.

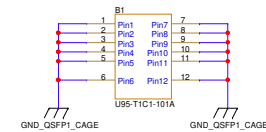
NOTE 2: QSFP 100-ohm termination is implemented via the FPGA on-chip termination.

NOTE 3: DC blocking capacitors are in the module for RX and TX.

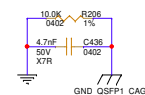
NOTE 4: 1uH inductors should have a DC Resistance of less than 0.1-ohm.



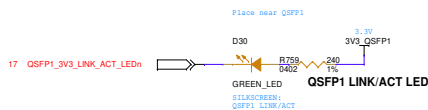
Place close to QSFP Connector



Amphenol U95-T1C1-101A

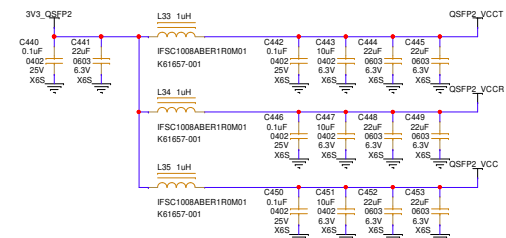
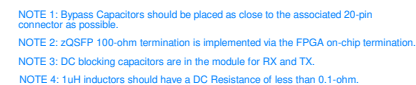


Amphenol FS1-K38-202A-A0

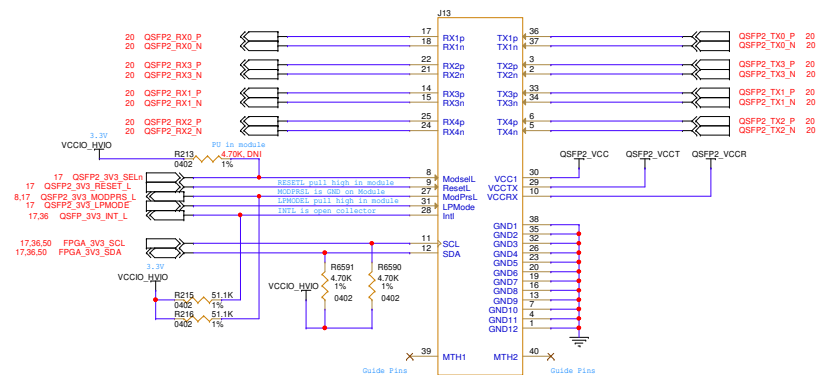


Altera Corporation, 161 Innovation Dr., San Jose CA 95134			
© 2015 Altera Corporation. All rights reserved.			
File	AGILEX 5 E-SERIES 905A PREMIUM DEV KIT (PRODUCTION)		
Size	Document Number	Rev	
C	160-0330715-A3	A3	
Date:	Monday, February 09, 2016	Sheet	38 of 67

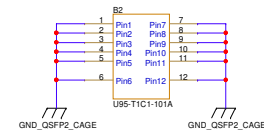
altera



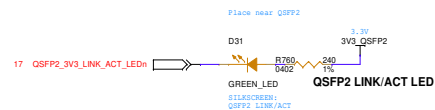
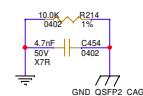
Place close to zQSFP Connector



Amphenol FS1-K38-20ZA-A0



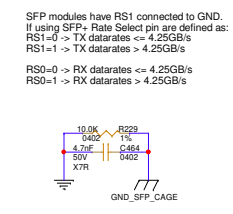
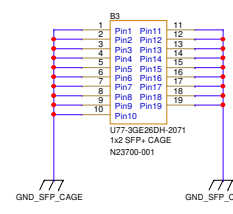
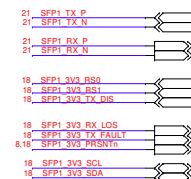
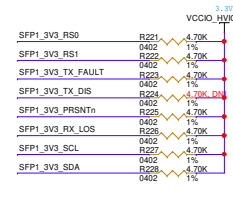
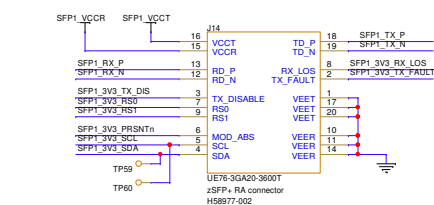
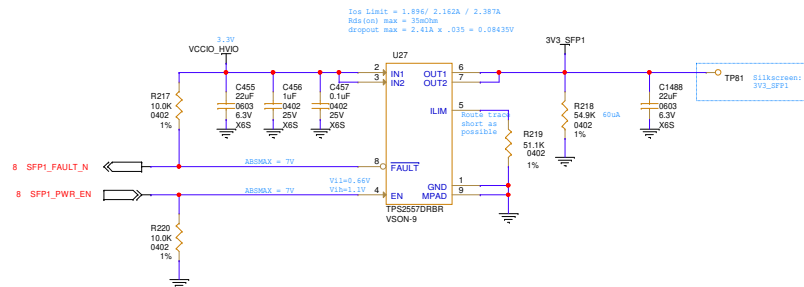
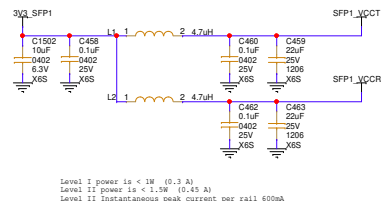
Amphenol U95-T1C1-101A



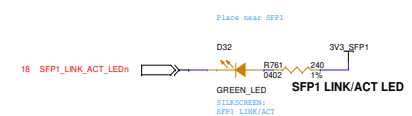
NOTE 1: Bypass Capacitors should be placed as close to the associated 20-pin connector as possible.

NOTE 2: Assuming that the SFP RD 100-ohm termination on the Host Board FPGA device will be implemented via the on-chip termination circuit.

NOTE 3: DC blocking capacitors are in the module for RX and TX.



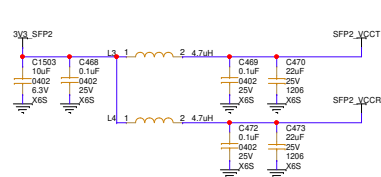
Optical (SFP+) Transceiver Cage 1x2



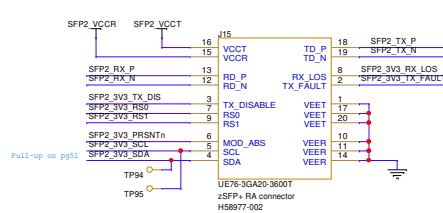
NOTE 1: Bypass Capacitors should be placed as close to the associated 20-pin connector as possible.

NOTE 2: Assuming that the SFP RD 100-ohm termination on the Host Board FPGA device will be implemented via the on-chip termination circuit.

NOTE 3: DC blocking capacitors are in the module for RX and TX.

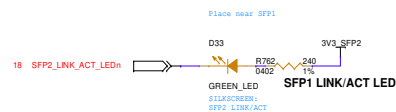


Level 1 power is < 1W (0.3 A)
Level 11 power is < 1.5W (0.43 A)
Level 11 Instantaneous peak current per rail 600mA



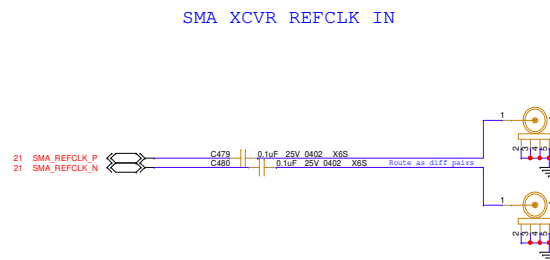
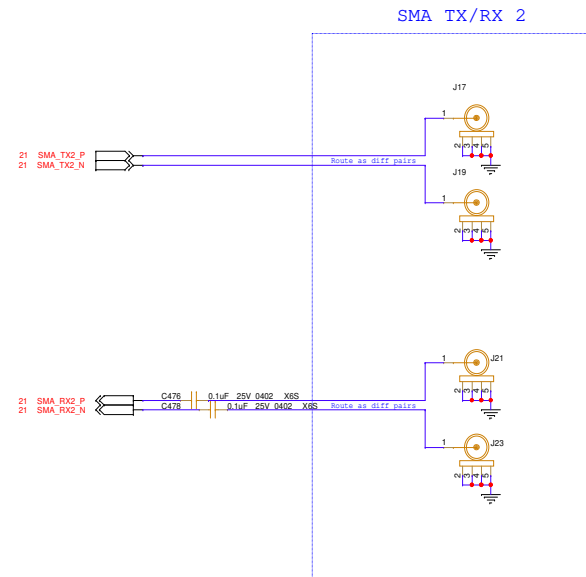
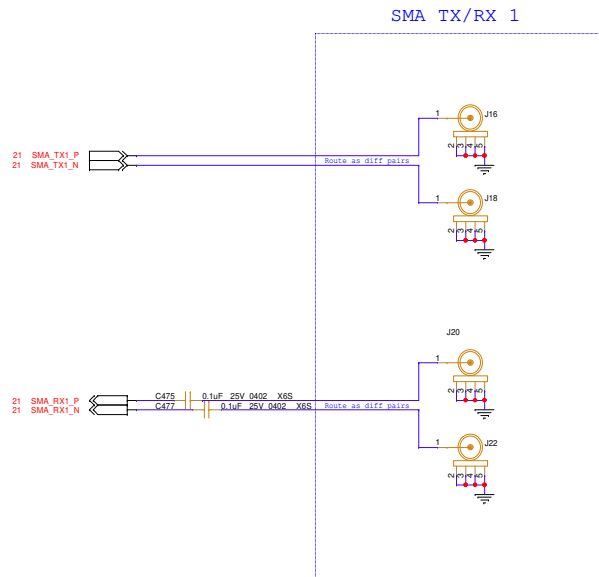
Optical (SFP+) Connector, case in page 38 B3

SFP modules have RS1 connected to GND.
If using SFP+ Rate Select pin are defined as:
RS1=0 -> TX data rates <= 4.25GB/s
RS1=1 -> TX data rates > 4.25GB/s
RS0=0 -> RX data rates <= 4.25GB/s
RS0=1 -> RX data rates > 4.25GB/s



altera

Altera Corporation, 161 Innovation Dr., San Jose CA 95134			
© 2008 Altera Corporation. All rights reserved.			
File	AGILEX 5 E-SERIES 905A PREMIUM DEV KIT (PRODUCTION)		
Size	Document Number	Rev	
C	160-0300713-A3	A3	
Date:	Monday, February 09, 2008	Sheet	39 of 67



altera

Altera Corporation, 101 Innovation Dr., San Jose CA 95134			
© 2015 Altera Corporation. All rights reserved.			
File AGILEX 5 E-SERIES 905A PREMIUM DEV KIT (PRODUCTION)			
Size C	Document Number 160-0330713-A3	Rev A3	
Date: Monday, February 09, 2016	Sheet 40	of 67	

FMC+ Connector (Vita 57.4) Complete Pin Table

FMC Connector (Vita 57.1) Complete Pin Table

14 x 40	M	L	K	J	H	G	F	E	D	C	B	A	Z	Y
1	GND	RES1	VREF_B_M2C	GND	VREF_A_M2C	GND	PG_M2C	GND	PG_C2M	GND	CLK_DIR	GND	HSPC_PRSNT_M2C_L	GND
2	DP23_M2C_P	GND	GND	CLK3_BIDIR_P	PRSNM_M2C_L	CLK1_M2C_P	GND	HA01_P_CC	GND	DP0_C2M_P	GND	DP1_M2C_P	GND	DP23_C2M_P
3	DP23_M2C_N	GND	GND	CLK3_BIDIR_N	GND	CLK1_M2C_N	GND	HA01_N_CC	GND	DP0_C2M_N	GND	DP1_M2C_N	GND	DP23_C2M_N
4	GND	GBTCLK4_M2C_P	CLK2_BIDIR_P	GND	CLK0_M2C_P	GND	HA00_P_CC	GND	GBTCLK0_M2C_P	GND	DP9_M2C_P	GND	DP22_C2M_P	GND
5	GND	GBTCLK4_M2C_N	CLK2_BIDIR_N	GND	CLK0_M2C_N	GND	HA00_N_CC	GND	GBTCLK0_M2C_N	GND	DP9_M2C_N	GND	DP22_C2M_N	GND
6	DP22_M2C_P	GND	GND	HA03_P	GND	LA00_P_CC	GND	HA05_P	GND	DP0_M2C_P	GND	DP2_M2C_P	GND	DP21_C2M_P
7	DP22_M2C_N	GND	GND	HA03_N	GND	LA00_N_CC	GND	HA05_N	GND	DP0_M2C_N	GND	DP2_M2C_N	GND	DP21_C2M_N
8	GND	GBTCLK3_M2C_P	HA02_N	GND	LA02_N	GND	HA04_N	GND	LA01_P_CC	GND	DP8_M2C_P	GND	DP20_C2M_P	GND
9	GND	GBTCLK3_M2C_N	HA02_P	GND	LA02_P	GND	HA04_P	GND	LA01_N_CC	GND	DP8_M2C_N	GND	DP20_C2M_N	GND
10	DP21_M2C_P	GND	HA06_P	HA07_N	LA04_P	LA03_N	HA08_P	HA09_N	GND	LA06_P	GND	DP3_M2C_P	GND	DP10_M2C_P
11	DP21_M2C_N	GND	HA06_N	HA07_P	LA04_N	LA03_P	HA08_N	HA09_P	GND	LA06_N	GND	DP3_M2C_N	GND	DP10_M2C_N
12	GND	GBTCLK2_M2C_P	GND	HA11_P	GND	LA08_P	GND	HA13_P	GND	LA05_N	GND	DP7_M2C_P	GND	DP11_M2C_P
13	GND	GBTCLK2_M2C_N	GND	HA11_N	GND	LA08_N	GND	HA13_N	GND	LA05_P	GND	DP7_M2C_N	GND	DP11_M2C_N
14	DP20_M2C_P	GND	HA10_P	HA11_N	LA07_P	LA08_N	HA12_P	HA13_N	GND	LA09_P	LA10_P	DP4_M2C_P	GND	DP12_M2C_P
15	DP20_M2C_N	GND	HA10_N	HA11_P	LA07_N	LA08_P	HA12_N	HA13_P	GND	LA09_N	LA10_N	DP4_M2C_N	GND	DP12_M2C_N
16	GND	SYNC_C2M_P	HA17_P_CC	HA14_N	LA11_P	LA12_N	HA15_P	HA16_N	GND	LA09_N	LA10_N	DP6_M2C_P	GND	DP13_M2C_P
17	GND	SYNC_C2M_N	HA17_N_CC	HA14_P	LA11_N	LA12_P	HA15_N	HA16_P	GND	LA09_P	LA10_P	DP6_M2C_N	GND	DP13_M2C_N
18	DP14_C2M_P	GND	GND	HA18_P	GND	LA16_P	GND	HA20_P	LA13_N	LA14_P	GND	DP5_M2C_P	GND	DP14_M2C_P
19	DP14_C2M_N	GND	GND	HA18_N	GND	LA16_N	GND	HA20_N	LA13_P	LA14_N	GND	DP5_M2C_N	GND	DP14_M2C_N
20	GND	REFCLK_C2M_P	HA21_P	HA18_N	LA15_P	LA16_N	HA19_P	HA20_N	GND	LA14_N	GND	DP5_M2C_N	GND	DP14_M2C_N
21	GND	REFCLK_C2M_N	HA21_N	HA18_P	LA15_N	LA16_P	HA19_N	HA20_P	GND	LA14_P	GND	DP5_M2C_P	GND	DP14_M2C_P
22	DP15_C2M_P	GND	HA23_P	HA22_N	LA19_P	LA20_N	HB02_P	HB03_N	GND	LA18_P_CC	GND	DP1_C2M_P	GND	DP15_M2C_P
23	DP15_C2M_N	GND	HA23_N	HA22_P	LA19_N	LA20_P	HB02_N	HB03_P	GND	LA18_N_CC	GND	DP1_C2M_N	GND	DP15_M2C_N
24	GND	REFCLK_M2C_P	GND	HB01_P	GND	LA22_P	GND	HB05_P	LA23_N	GND	DP9_C2M_P	GND	DP10_C2M_P	GND
25	GND	REFCLK_M2C_N	GND	HB01_N	GND	LA22_N	GND	HB05_N	LA23_P	GND	DP9_C2M_N	GND	DP10_C2M_N	GND
26	DP16_C2M_P	GND	HB00_N_CC	GND	LA21_N	GND	HB04_N	HB05_P	GND	LA26_P	GND	DP2_C2M_P	GND	DP11_C2M_P
27	DP16_C2M_N	GND	HB00_P_CC	GND	LA21_P	GND	HB04_P	HB05_N	GND	LA26_N	GND	DP2_C2M_N	GND	DP11_C2M_N
28	GND	SYNC_M2C_P	HB06_P_CC	HB07_N	LA24_P	LA25_N	HB08_P	HB09_N	GND	LA27_N	GND	DP8_C2M_P	GND	DP12_C2M_P
29	GND	SYNC_M2C_N	HB06_N_CC	HB07_P	LA24_N	LA25_P	HB08_N	HB09_P	GND	LA27_P	GND	DP8_C2M_N	GND	DP12_C2M_N
30	DP17_C2M_P	GND	GND	HB11_P	GND	LA29_P	GND	HB13_P	TDI	SCL	GND	DP3_C2M_P	GND	DP13_C2M_P
31	DP17_C2M_N	GND	GND	HB11_N	GND	LA29_N	GND	HB13_N	TDO	SDA	GND	DP3_C2M_N	GND	DP13_C2M_N
32	GND	RES2	HB10_N	GND	LA28_N	GND	HB12_N	GND	3P3VAUX	GND	DP7_C2M_P	GND	DP16_M2C_P	GND
33	GND	RES3	HB10_P	GND	LA28_P	GND	HB12_P	GND	3P3VAUX	GND	DP7_C2M_N	GND	DP16_M2C_N	GND
34	DP18_C2M_P	GND	HB14_P	HB15_N	LA30_P	LA31_N	HB16_P	HB19_N	TRST_L	GA0	GND	DP4_C2M_P	GND	DP17_M2C_P
35	DP18_C2M_N	GND	HB14_N	HB15_P	LA30_N	LA31_P	HB16_N	HB19_P	TRST_N	GA0	GND	DP4_C2M_N	GND	DP17_M2C_N
36	GND	12P0V	GND	HB18_P	GND	LA33_P	GND	HB21_P	3P3V	GND	DP6_C2M_P	GND	DP18_M2C_P	GND
37	GND	12P0V	GND	HB18_N	GND	LA33_N	GND	HB21_N	3P3V	GND	DP6_C2M_N	GND	DP18_M2C_N	GND
38	DP19_C2M_P	GND	HB17_P_CC	HB18_N	LA32_P	LA33_N	HB20_P	HB21_N	GND	3P3V	GND	DP5_C2M_P	GND	DP19_M2C_P
39	DP19_C2M_N	GND	HB17_N_CC	HB18_P	LA32_N	LA33_P	HB20_N	HB21_P	GND	3P3V	GND	DP5_C2M_N	GND	DP19_M2C_N
40	GND	12P0V	VIO_B_M2C	GND	VADJ	GND	VADJ	GND	3P3V	GND	RES0	GND	3P3V	GND

FMC+ Voltage Specification

Voltage Supply	Allowable Voltage Range	No Pins	Max Amps	Tolerance	Max Capacitive Load
VADJ	0 - 3.3V	4	4	+/- 5%	1000 uF
VIO_B_M2C	0 - VADJ	2	1.15	+/- 5%	500 uF
VREF_A_M2C	0 - VADJ	1	1mA	+/- 2%	10 uF
VREF_B_M2C	0 - VIO_B_M2C	1	1mA	+/- 2%	10 uF
3P3VAUX	3.3V	1	20mA	+/- 5%	150 uF
3P3V	3.3V	5	3	+/- 5%	1000 uF
12P0V	12V	5	1	+/- 5%	1000 uF

Supported FMC Cards:

1. FMC+ PCIe Gen4 x16 RootPort Card (HighTech Global)
- ** Note: Only support PCIe Gen4 x4 **
2. FMC+ HDMI 2.1 Card Rev9 (Bitec)
3. FMC+ DP 1.4 Rev12 (Bitec)

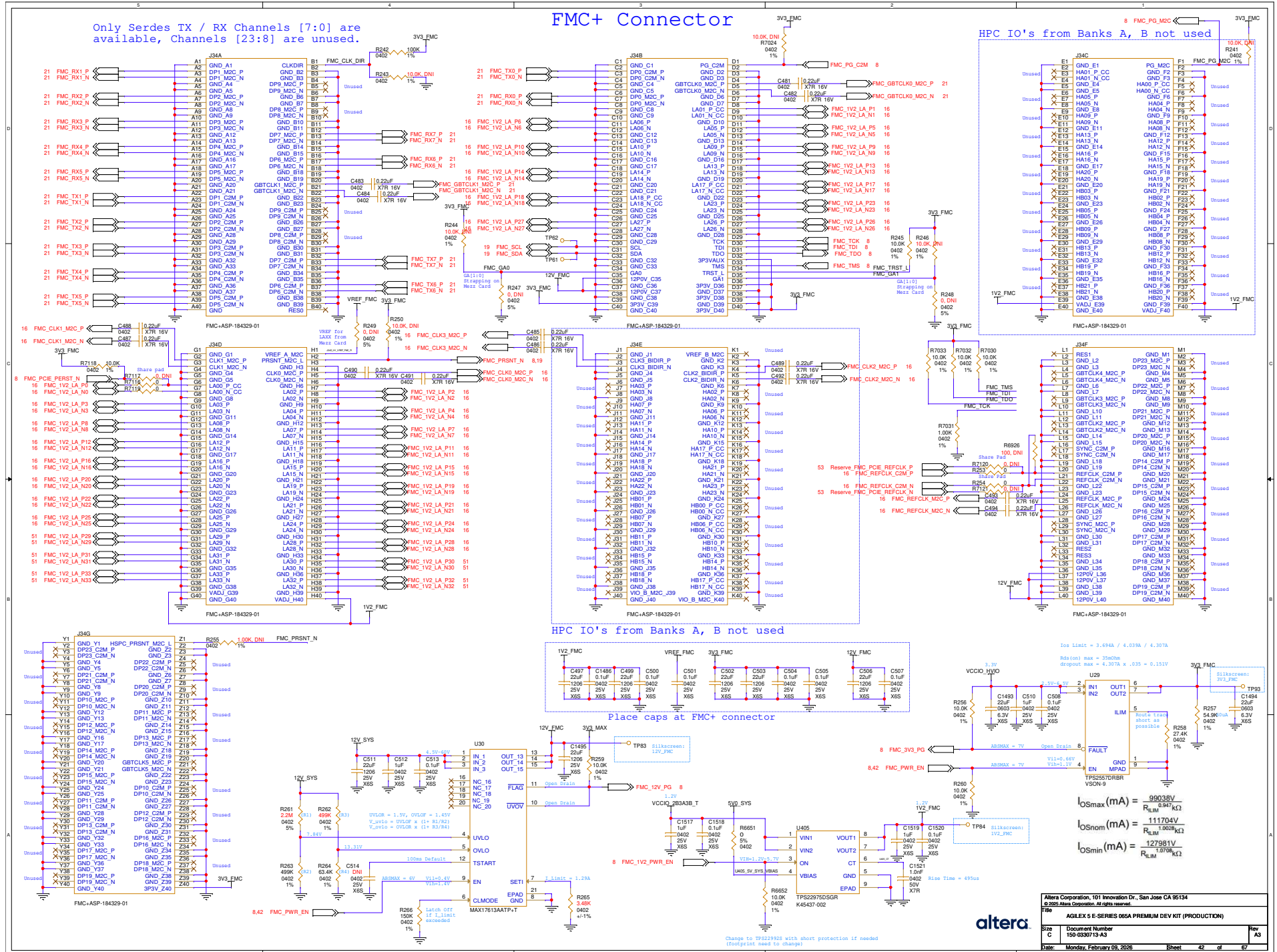
altera

Altera Corporation, 161 Innovation Dr., San Jose CA 95134			
© 2015 Altera Corporation. All rights reserved.			
File	AGILEX 5 E-SERIES 955A PREMIUM DEV KIT (PRODUCTION)		
Size	Document Number	160-0300715-A3	Rev A3
Date	Monday, February 09, 2016	Sheet	41 of 67

Only Serdes TX / RX Channels [7:0] are available, Channels [23:8] are unused.

FMC+ Connector

HPC IO's from Banks A, B not used



Altera Corporation, 161 Innovation Dr., San Jose CA 95134

Agilex 5 E-Series 905A Premium Dev Kit (Production)

File: AGLEX 5 E-Series 905A Premium Dev Kit (Production)

Size: 160-0300715-A3

Document Number: 160-0300715-A3

Date: Monday, February 09, 2016

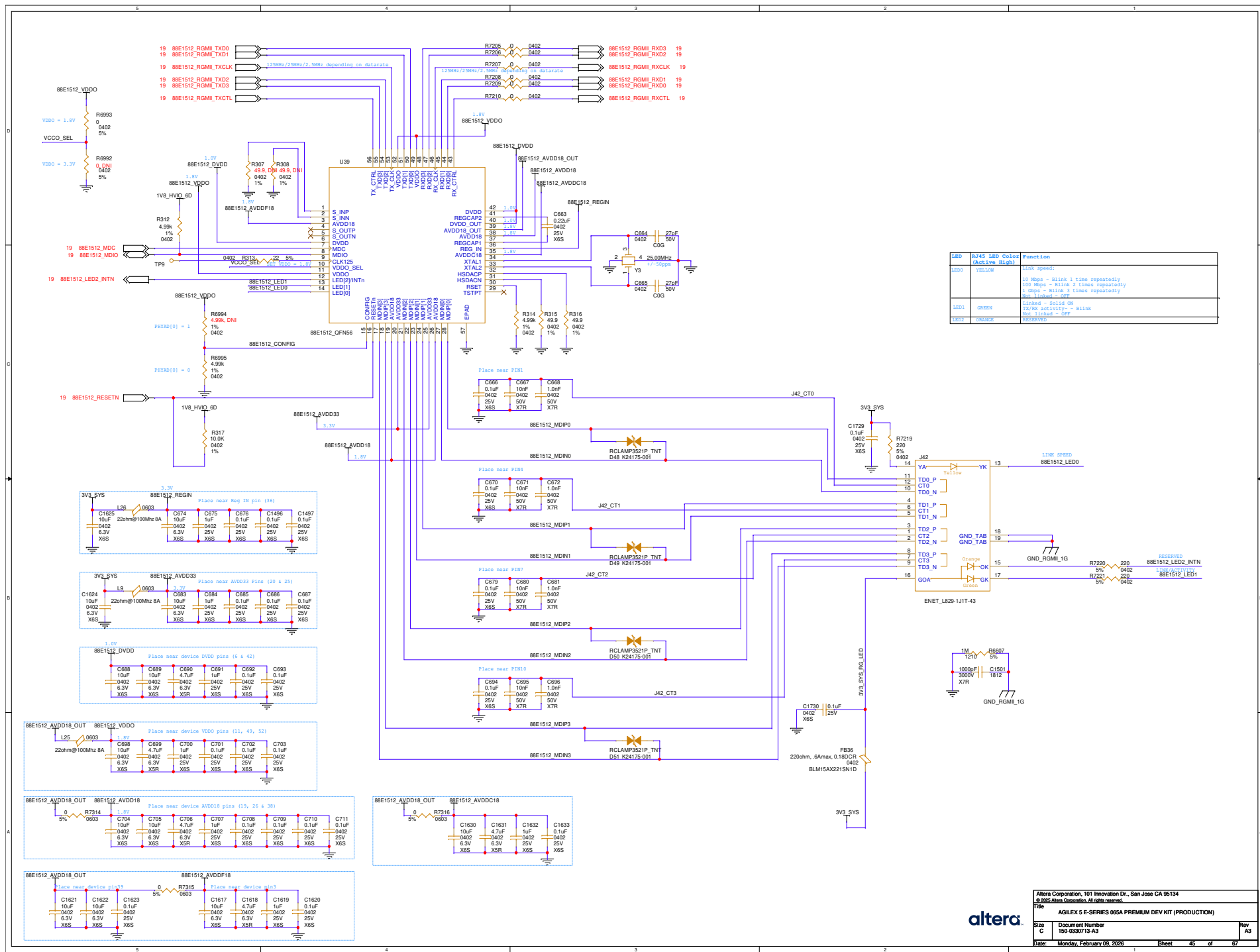
Sheet: 42 of 67

Change to TP229702 with short protection if needed (Copyright need to change)

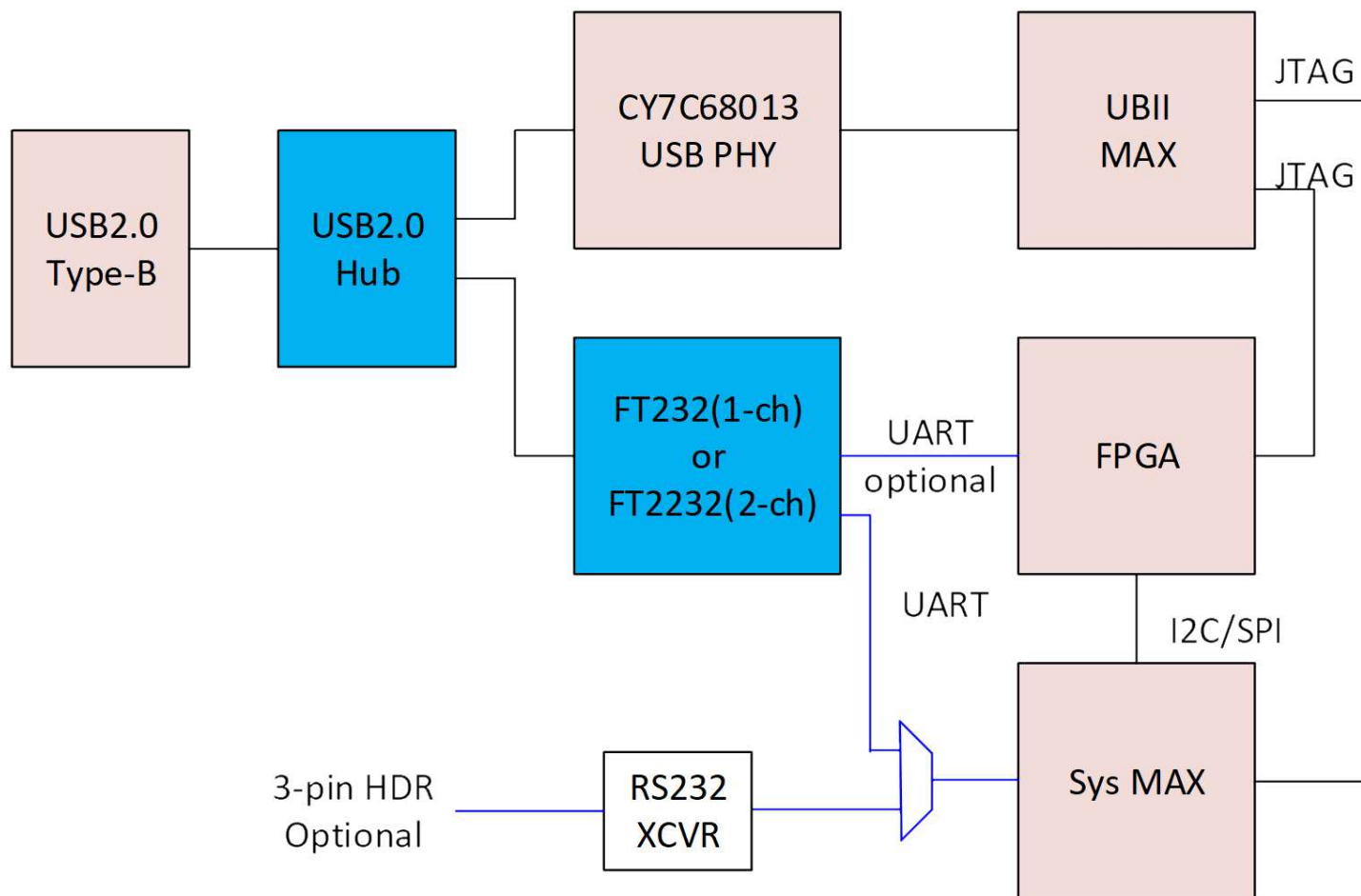
BLANK



Altera Corporation, 101 Innovation Dr., San Jose CA 95134			
© 2015 Altera Corporation. All rights reserved.			
Title			
AGILEX 5 E-SERIES 065A PREMIUM DEV KIT (PRODUCTION)			
Size	Document Number		Rev
C	150-0330713-A3		A3
Date:	Monday, February 09, 2016	Sheet	44 of 67

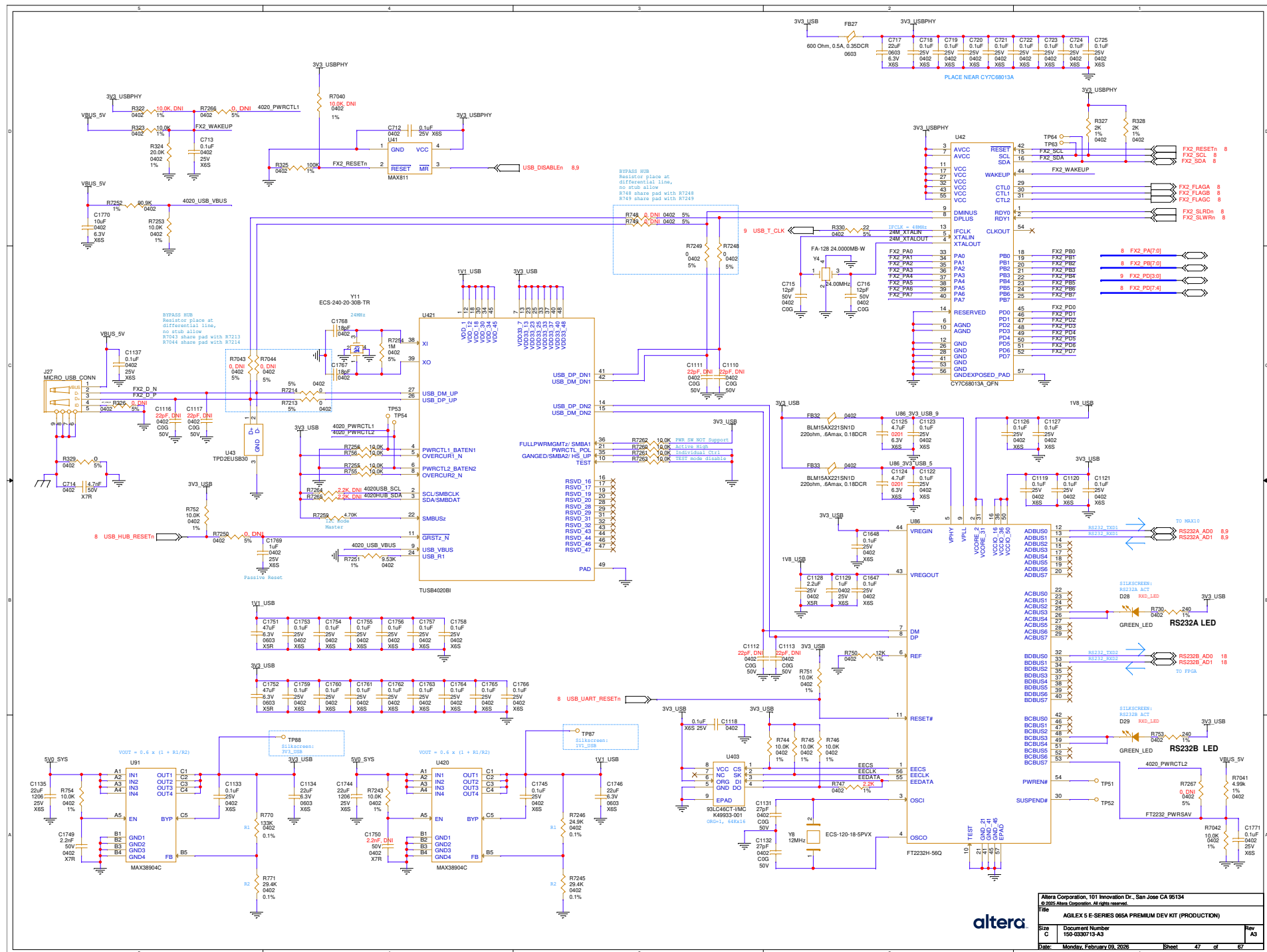


LED	4x4 LED Color (Active High)	Function
LED0	YELLOW	Link Speed: 10 Mbps - Blink 1 time repeatedly 100 Mbps - Blink 2 times repeatedly 1 Gbps - Blink 3 times repeatedly Not linked - OFF
LED1	GREEN	Linked - Solid ON 2x/8x Activity - Blink Not linked - OFF
LED2	ORANGE	RESERVED

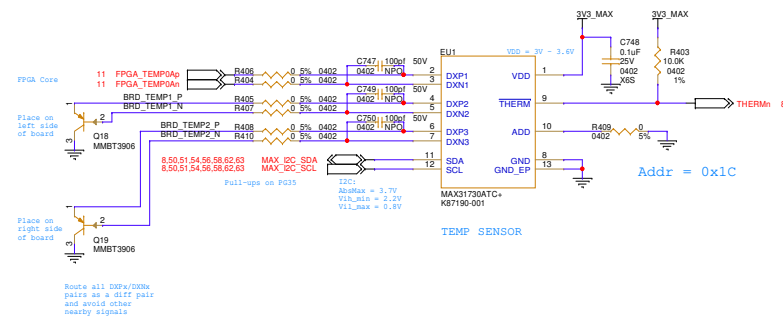
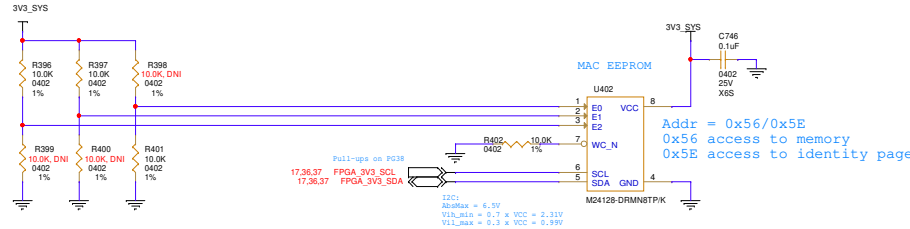
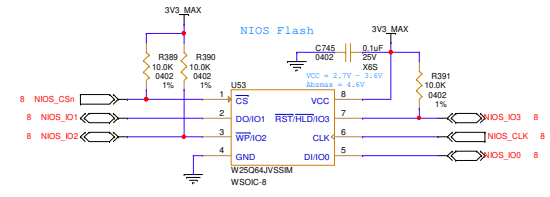
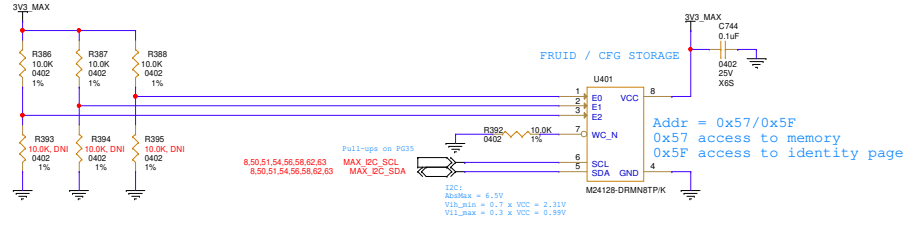


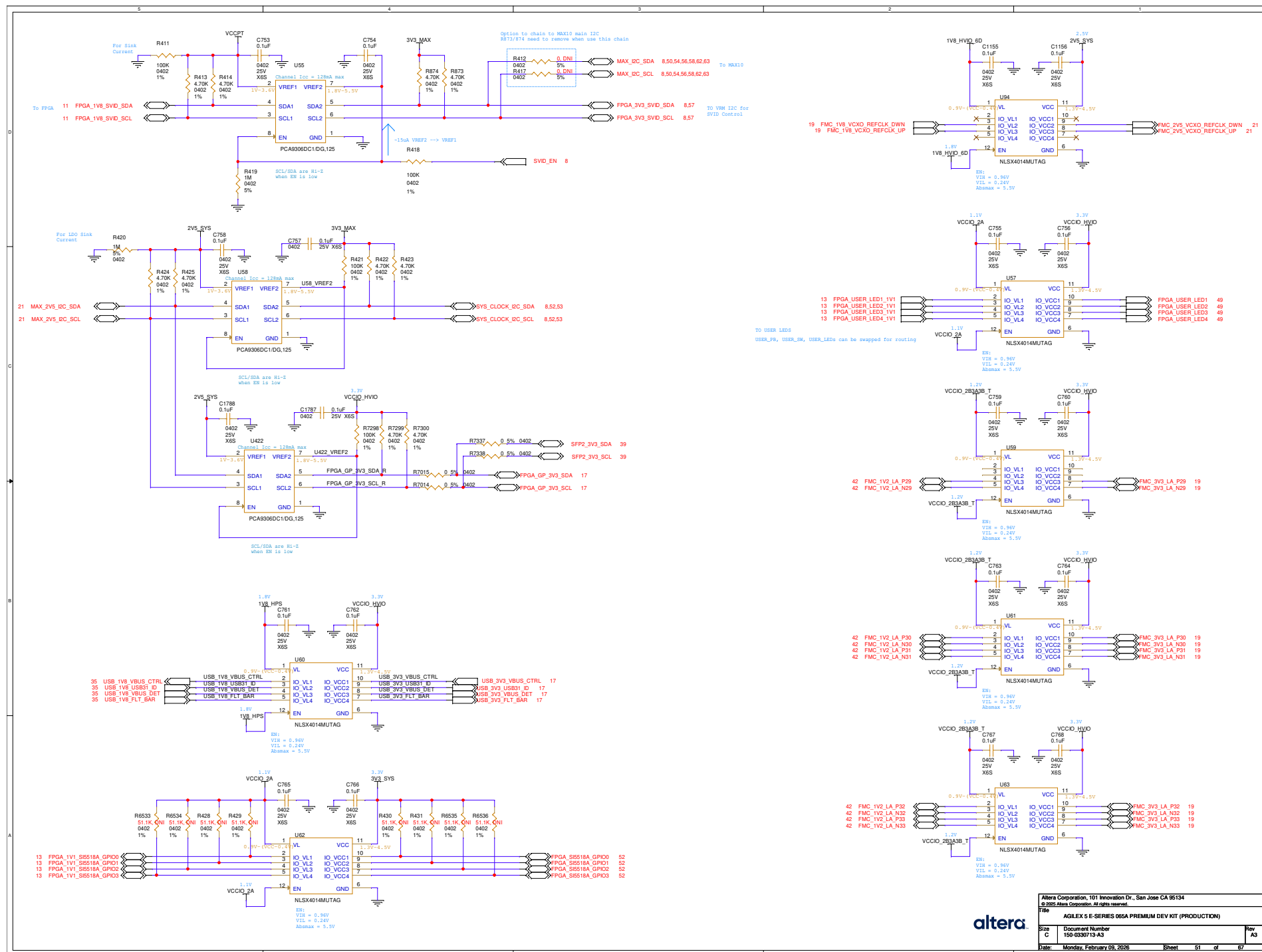
altera

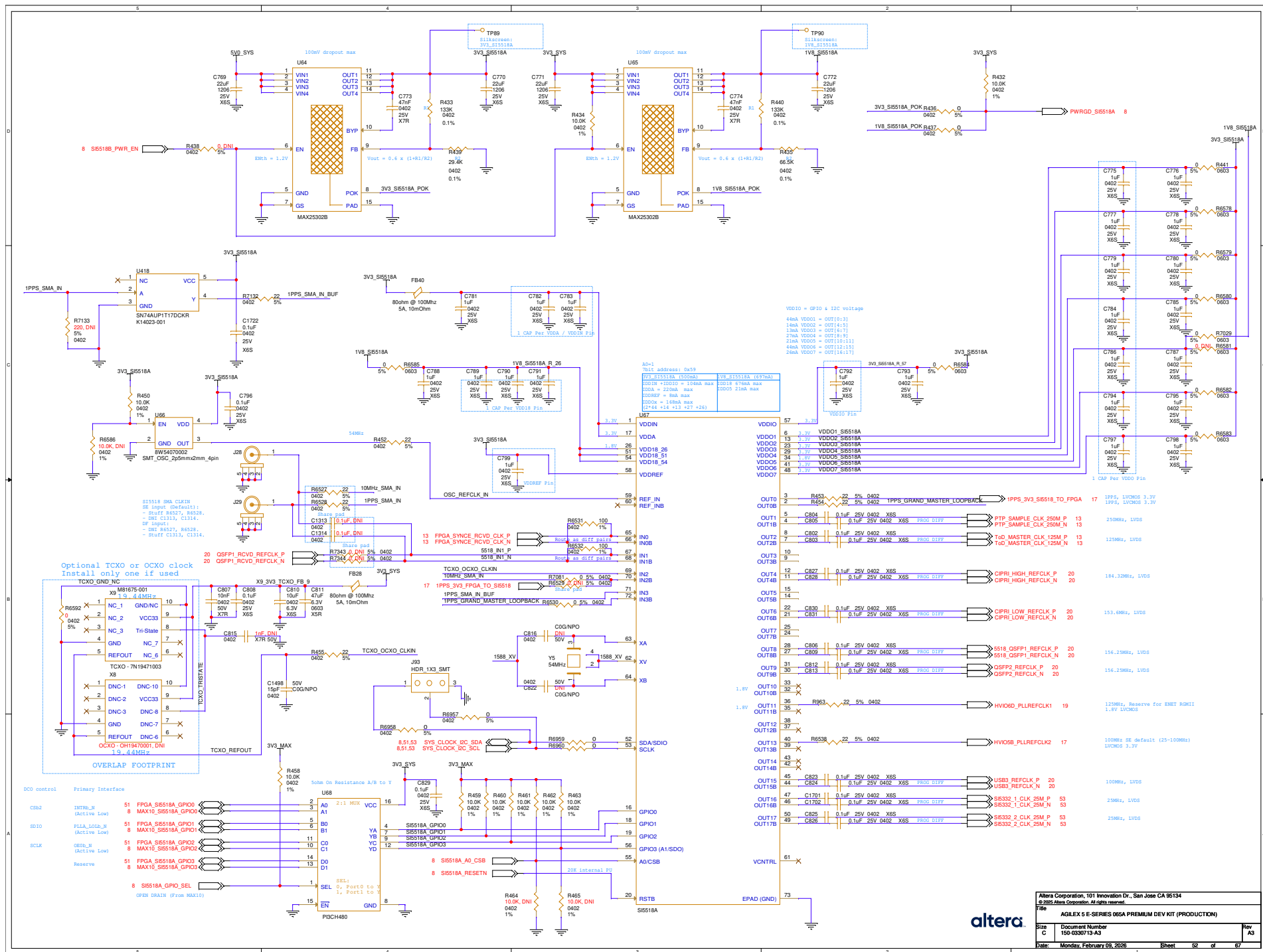
Altera Corporation, 101 Innovation Dr., San Jose CA 95134			
© 2015 Altera Corporation. All rights reserved.			
File: AGILEX 5 E-SERIES 955A PREMIUM DEV KIT (PRODUCTION)			
Size:	Document Number	Rev	
C	150-0330713-A3	A3	
Date:	Monday, February 09, 2015	Sheet	48 of 67









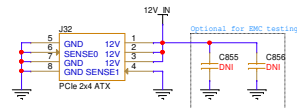


altera

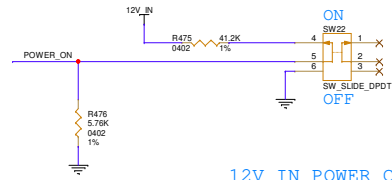
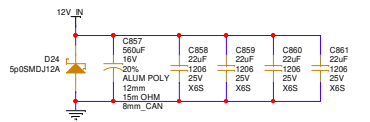
Altera Corporation, 161 Innovation Dr., San Jose CA 95134			
© 2015 Altera Corporation. All rights reserved.			
File	AGILEX 5 E-SERIES SOCA PREMIUM DEV KIT (PRODUCTION)		
Size	Document Number	160-0300713-A3	
C	Document Number	160-0300713-A3	
Date	Monday, February 09, 2015	Sheet	52 of 67



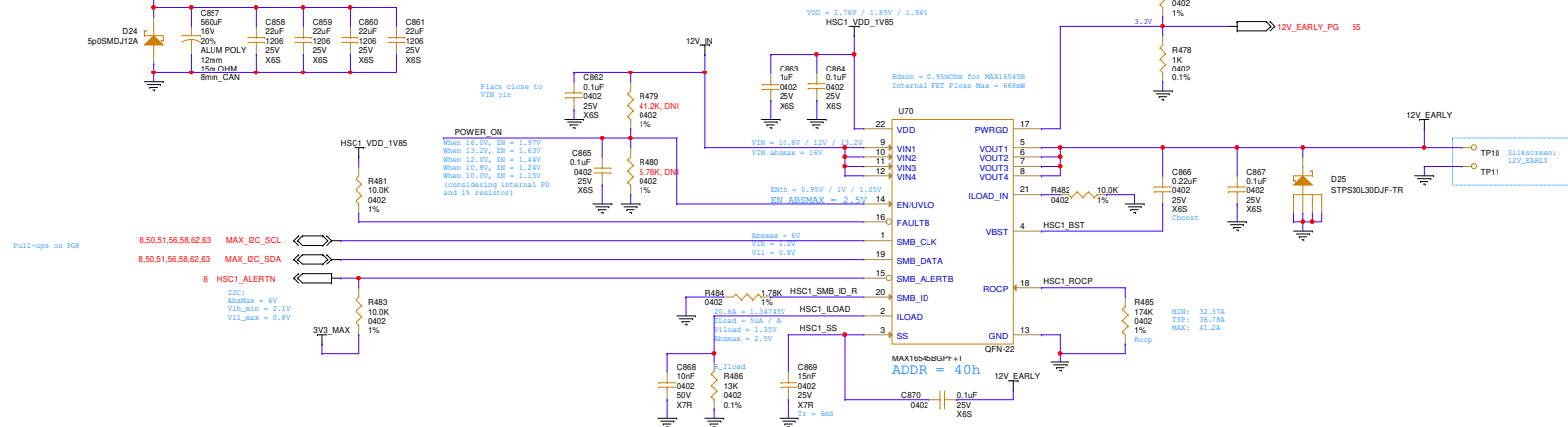
Power On Switch

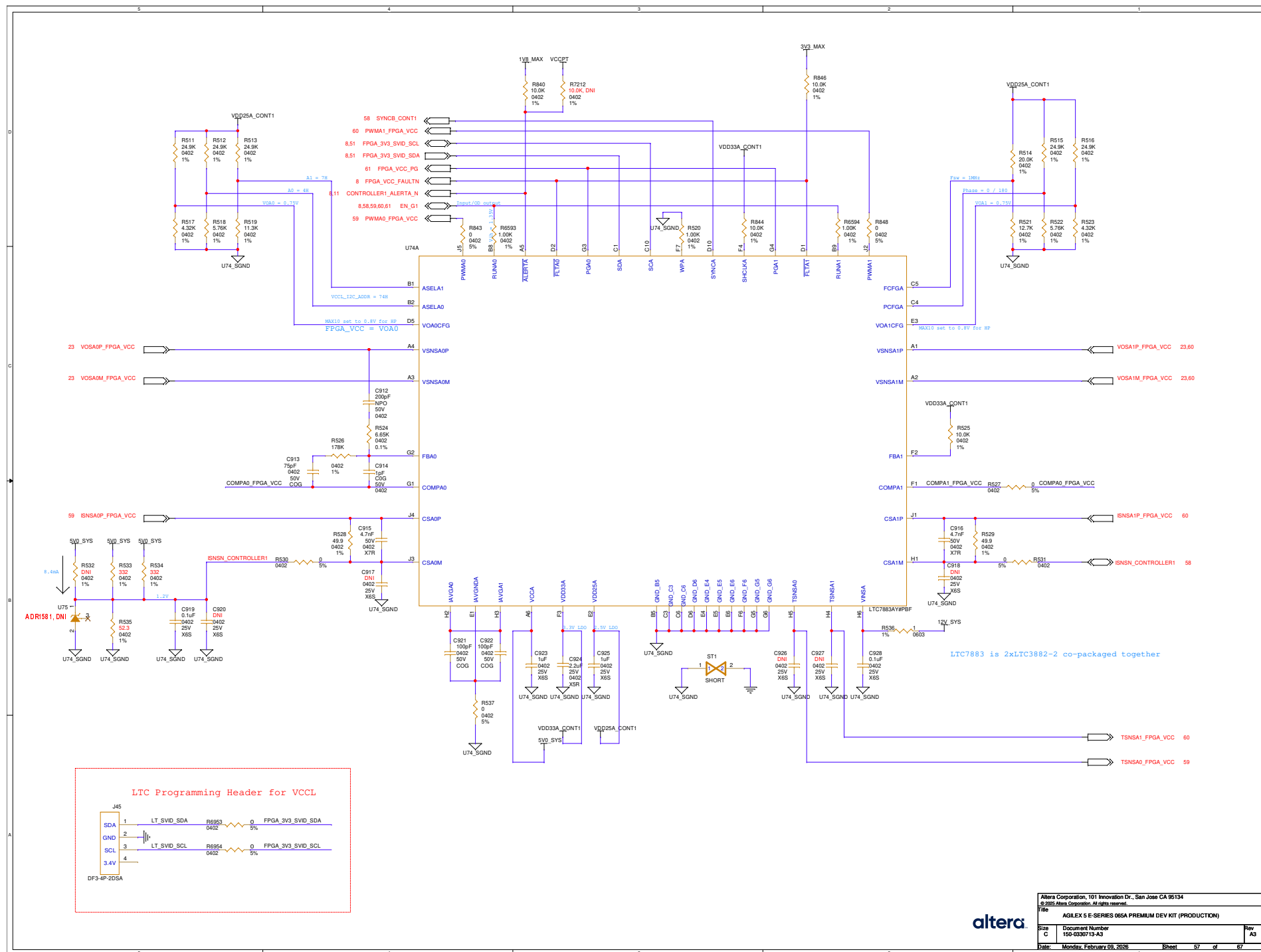


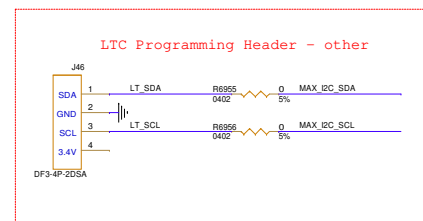
150W 2x4 ATX AUX
INPUT POWER
CONNECTOR



12V_IN POWER CONDITIONER
(MAX10 POWER DOMAIN)



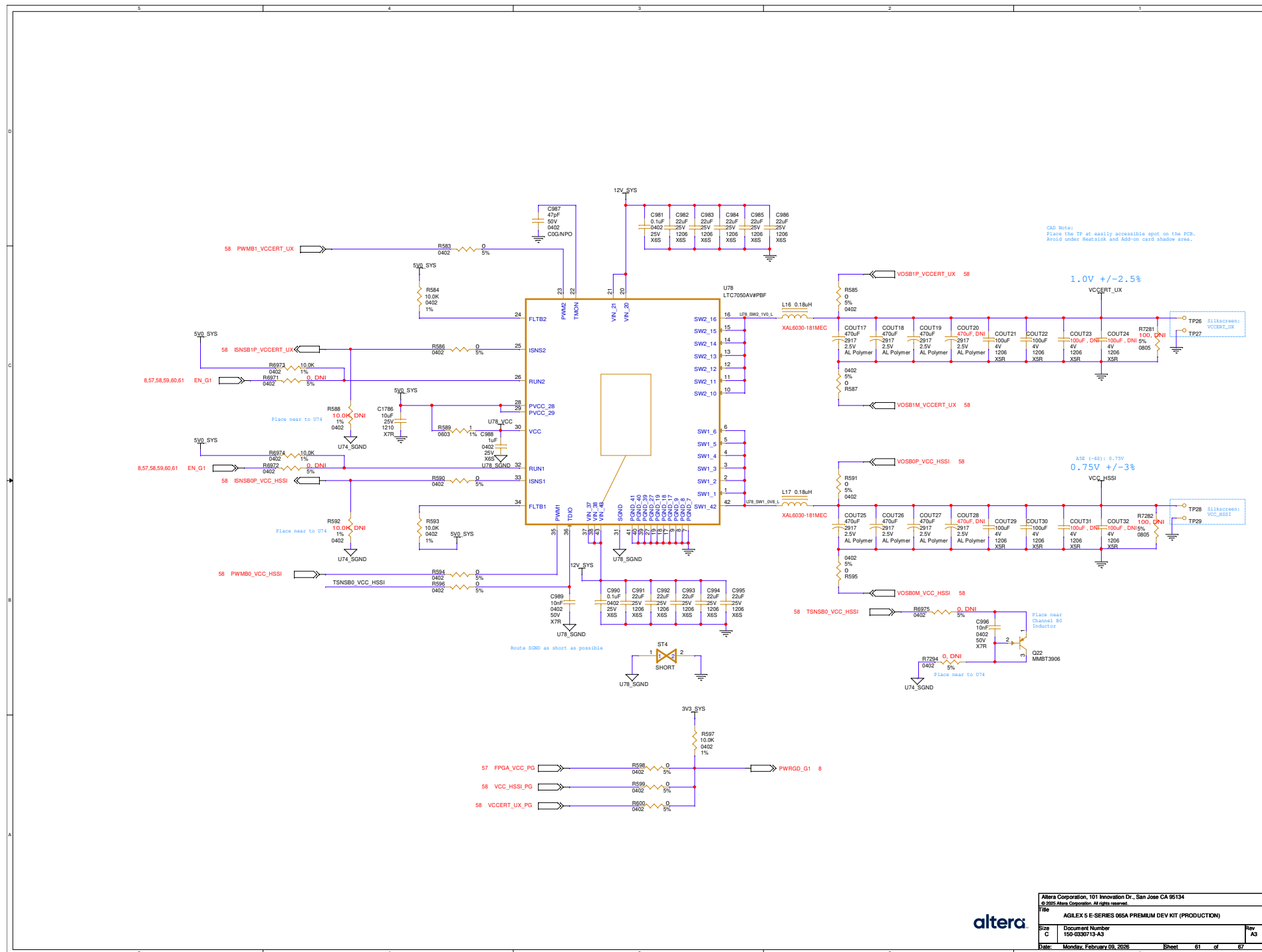




LTC7883 is 2xLTC3882-2 co-packaged together

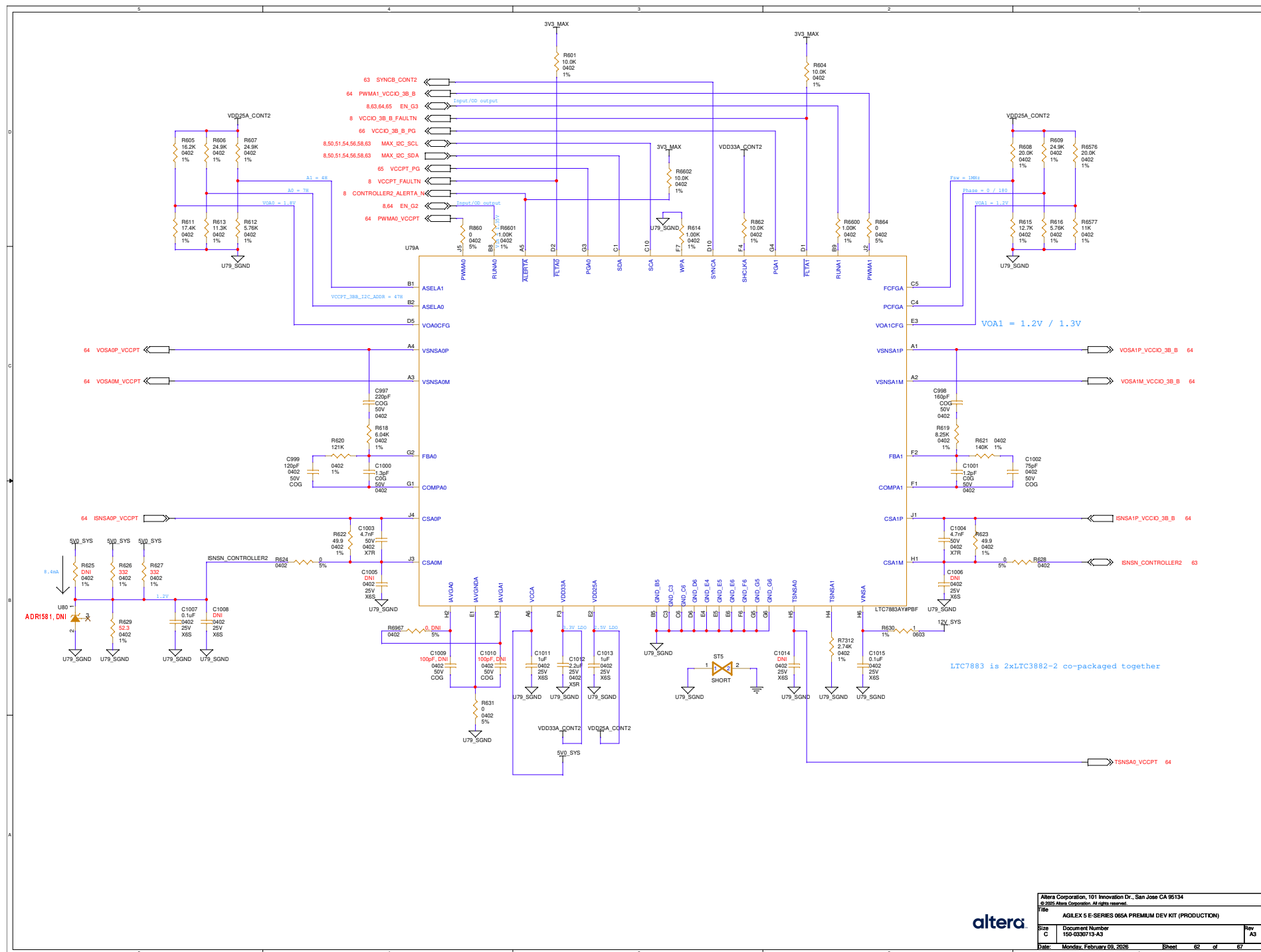
Altera Corporation, 101 Innovation Dr., San Jose CA 95134			
© 2015 Altera Corporation. All rights reserved.			
File	AGILEX 5 E-SERIES 905A PREMIUM DEV KIT (PRODUCTION)		
Size	Document Number	Rev	
C	160-0330713-A3	A3	
Date:	Monday, February 09, 2016	Sheet	58 of 67

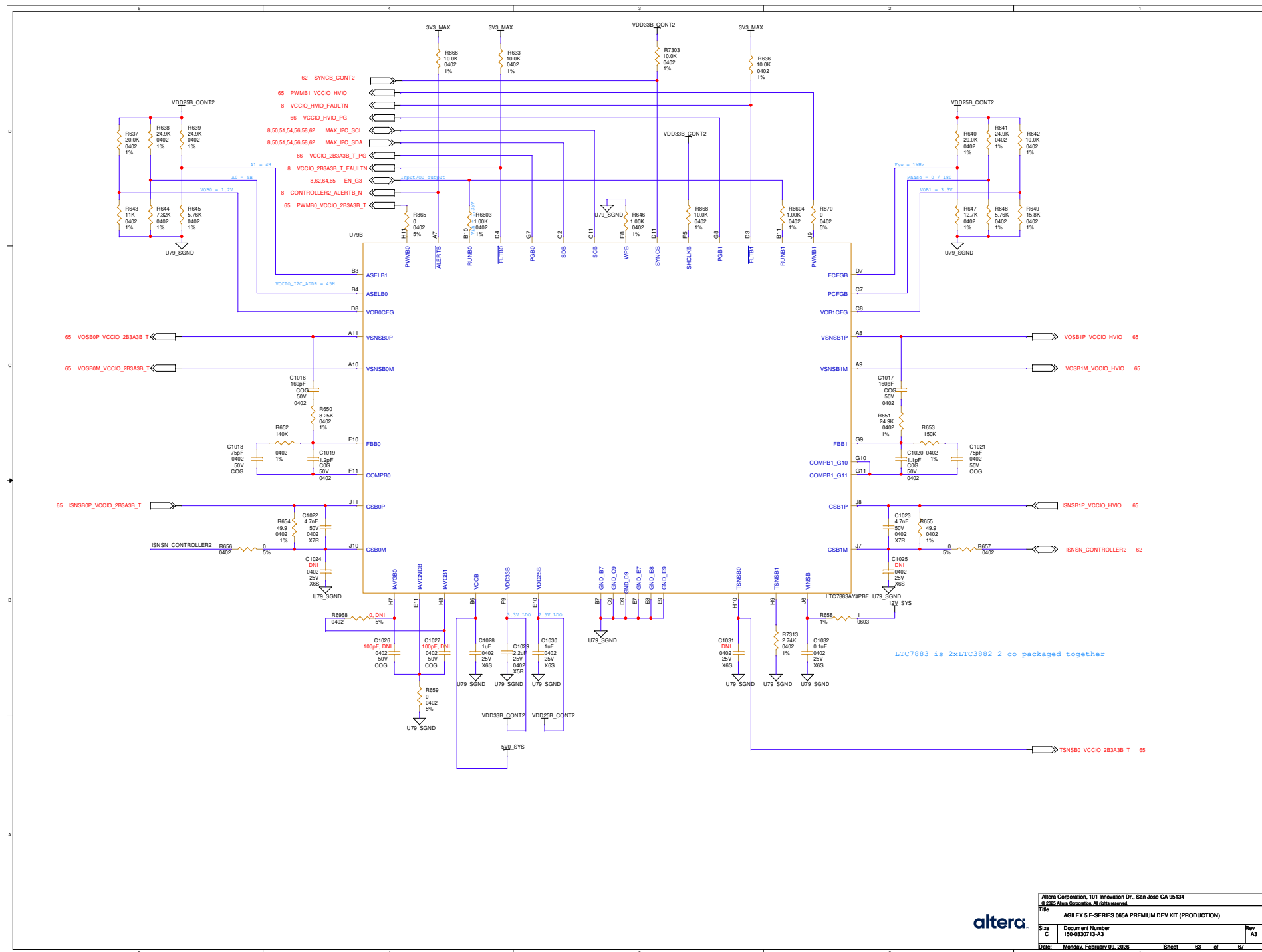
altera

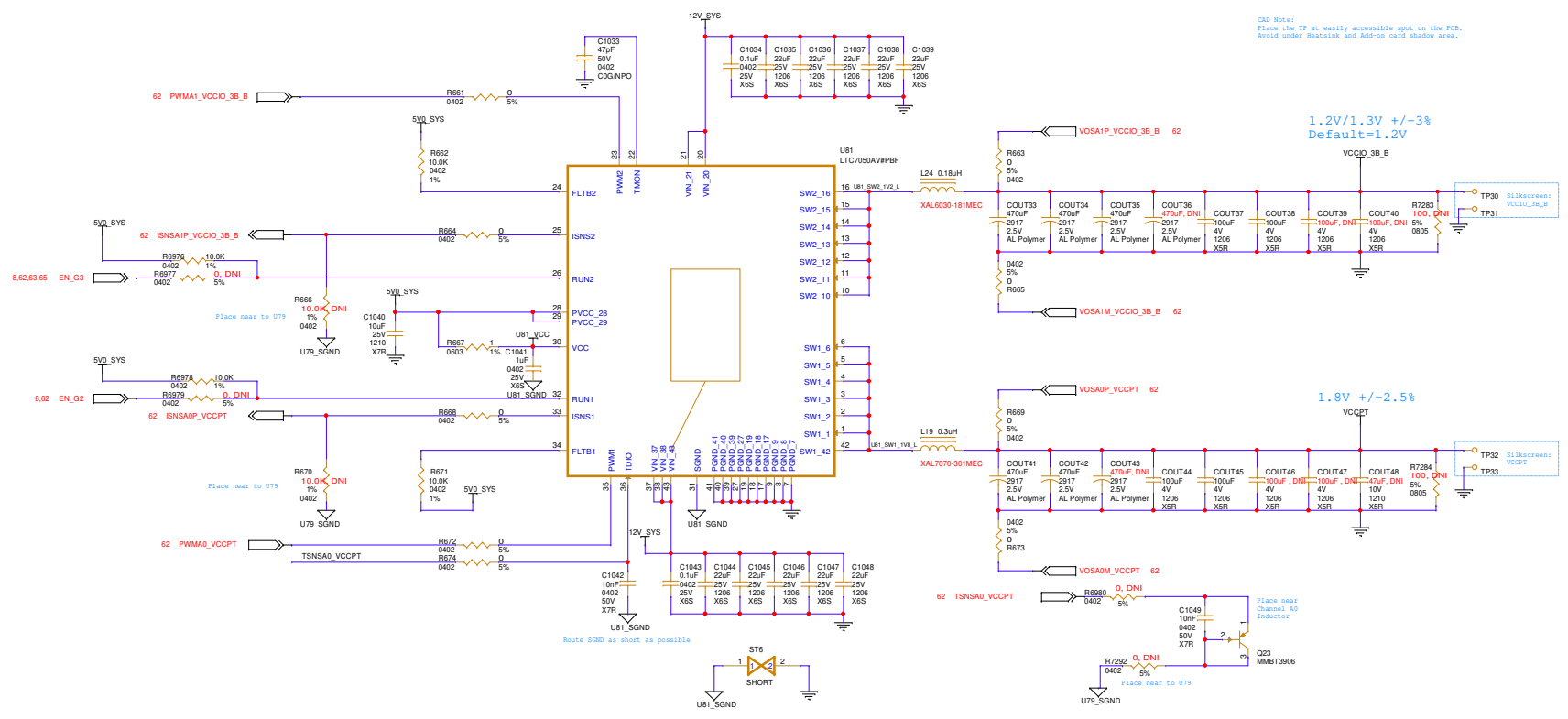


altera

Altera Corporation, 161 Innovation Dr., San Jose CA 95134			
© 2015 Altera Corporation. All rights reserved.			
File	AGILEX 5 E-SERIES 905A PREMIUM DEV KIT (PRODUCTION)		
Size	Document Number	160-0330715-A3	Rev
C			A3
Date:	Monday, February 09, 2016	Sheet	61 of 67







CAO Note:
Place the TP at easily accessible spot on the PCB.
Avoid under Heatsink and Add-on card shadow area.

1.2V/1.3V +/-3%
Default=1.2V

1.8V +/-2.5%

altera

Altera Corporation, 161 Innovation Dr., San Jose CA 95134			
© 2015 Altera Corporation. All rights reserved.			
File	AGILEX 5 E-SERIES 905A PREMIUM DEV KIT (PRODUCTION)		
Size	Document Number	Rev	
C	160-0330715-A3	A3	
Date:	Monday, February 09, 2016	Sheet	64 of 67

